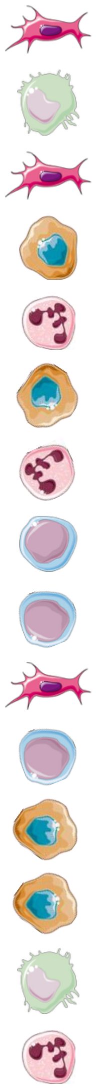
Introduction to domain identification and niche reconstruction

Ahmed Mahfouz

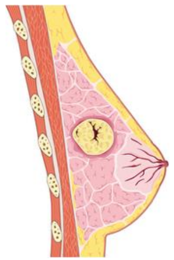
Human Genetics, Leiden University Medical Center
Pattern Recognition and Bioinformatics, TU Delft

Slides acknowledgements: Anna Schaar, Marco Varrone, Francesca Drummer

scRNA-seq



Spatial transcriptomics



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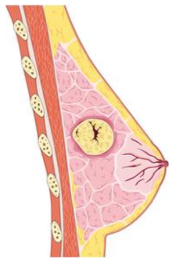

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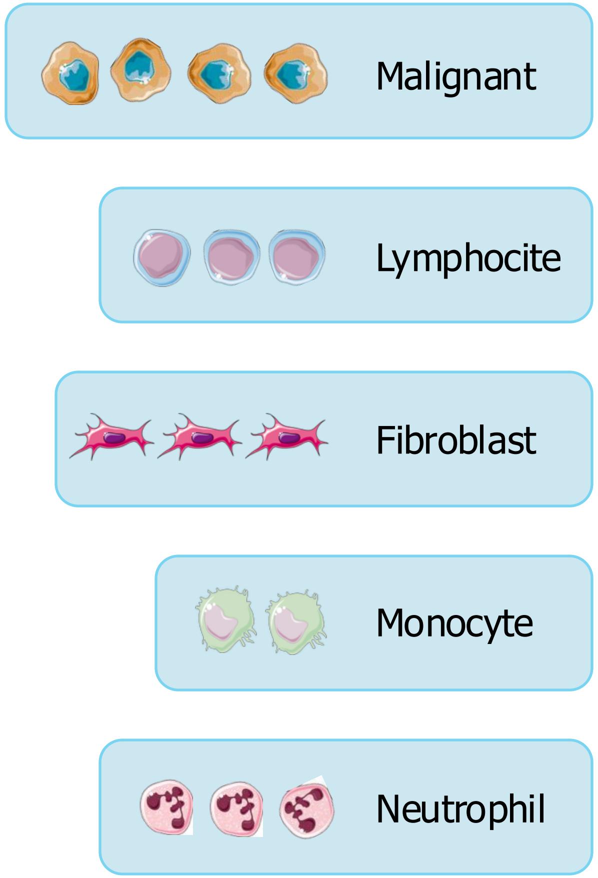

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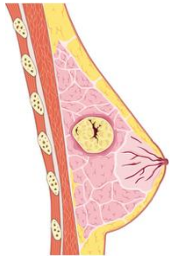

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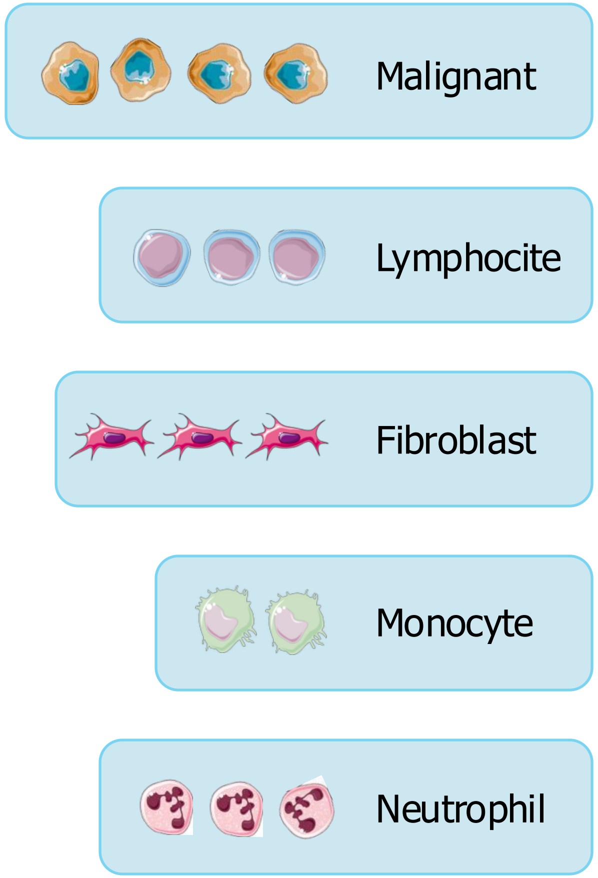

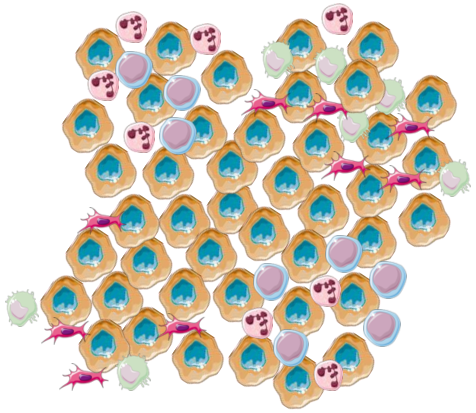

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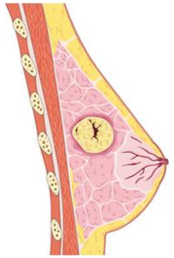

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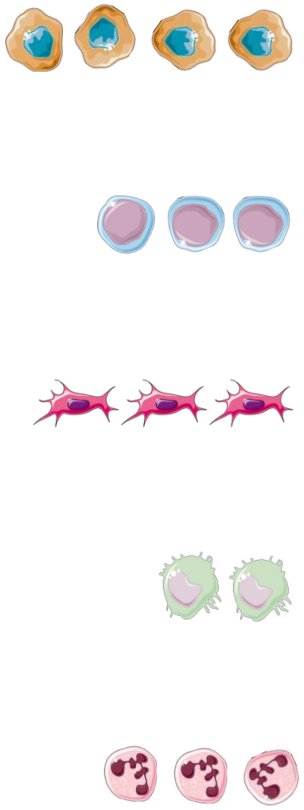


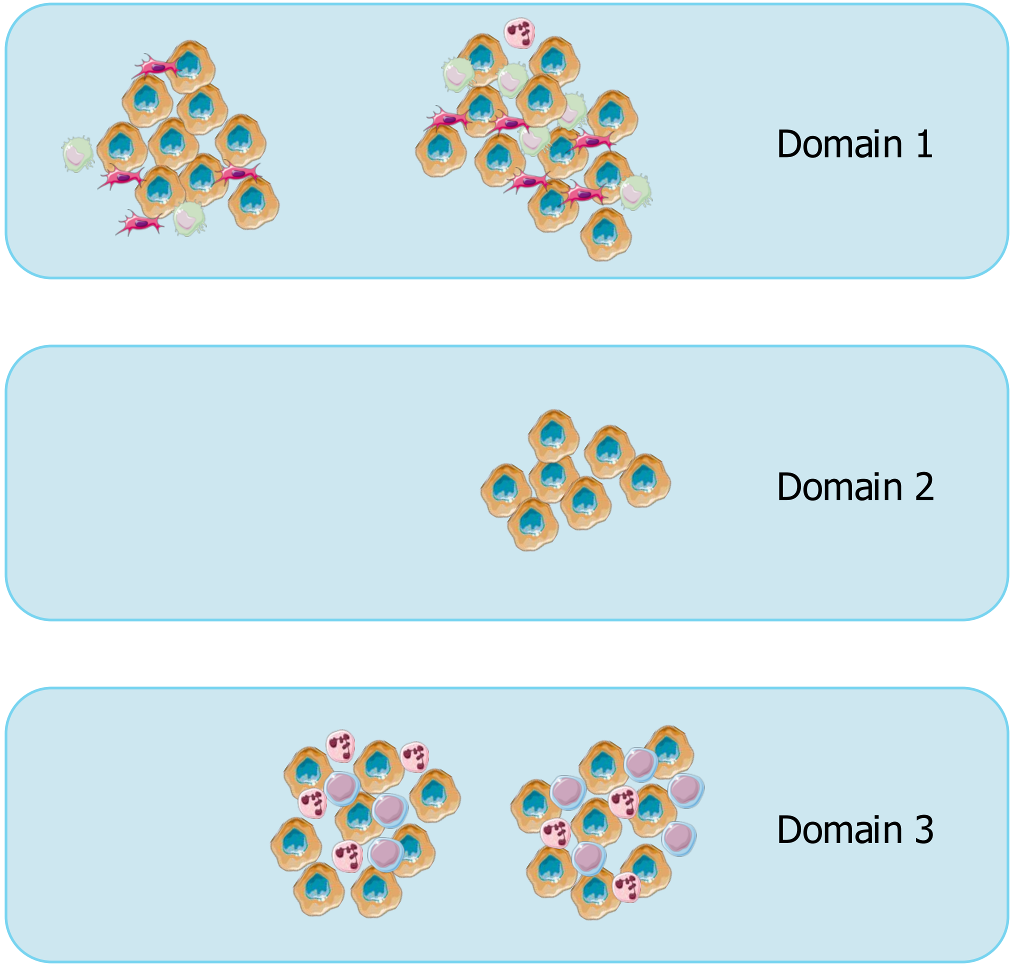
Malignant

Lymphocite

Fibroblast

Monocyte

Neutrophil



What are niches/domains/... in spatial omics?

Method | Open access | Published: 12 January 2024

Niche-DE: niche-differential gene expression analysis in spatial transcriptomics data identifies context-dependent cell-cell interactions

》 In our analyses, we define **a cell's niche as the cell-type composition** of its local neighborhood, where the **range** of the local neighborhood is determined by a kernel function.

12k Accesses | 43 Altmetric | Metrics

Brief Communication | Open access | Published: 27 October 2022

Modeling intercellular communication in tissues using spatial graphs of cells

David S. Fischer, Anna C. Schaar & Fabian J. Theis

Nature Biotechnology 41, 332–336 (2023) | Cite this article

》 **Spatial graphs** of cells are based on segmentation masks of cells in spatial molecular profiling data. Resolution is the **radius of neighborhood** used to define a **niche**.

Article | Open access | Published: 02 April 2024

The covariance environment defines cellular niches for spatial inference

Doron Haviv, Ján Remšík, Mohamed Gatie, Catherine Snopkowski, Meril Takizawa, Nathan Pereira, John Bashkin, Stevan Jovanovich, Tal Nawy, Ronan Chaligne, Adrienne Boire, Anna-Katerina Hadjantonakis &

》 The local neighborhood, or **niche**, of a cell is a useful resolution for **defining cell interactions**; it may represent **functional anatomical subunits** (such as stem cell niches) and is a basis for **identifying larger spatial patterns**.

Quantitative characterization of cell niches in spatial atlases

Sebastian Birk, Irene Bonafonte-Pardàs, Adib Miraki Feriz, Adam Boxall, Eneritz Agirre, Fani Memi, Anna Maguza, Rong Fan, Gonçalo Castelo-Branco, Fabian J. Theis, Omer Ali Bayraktar,

》 Identifying niches and quantifying their functions can help to **unravel important insights** into tissue architecture and **spatial biomarkers**, which can be harnessed to create novel diagnostic tools, identify new drug targets, and develop targeted therapies.

Nicheformer: a foundation model for single-cell and spatial omics

Anna C. Schaar, Alejandro Tejada-Lapuerta, Giovanni Palla
Mariia Minaeva, Larsen Vornholz, Leander Dony, Francesca
Fabian J. Theis

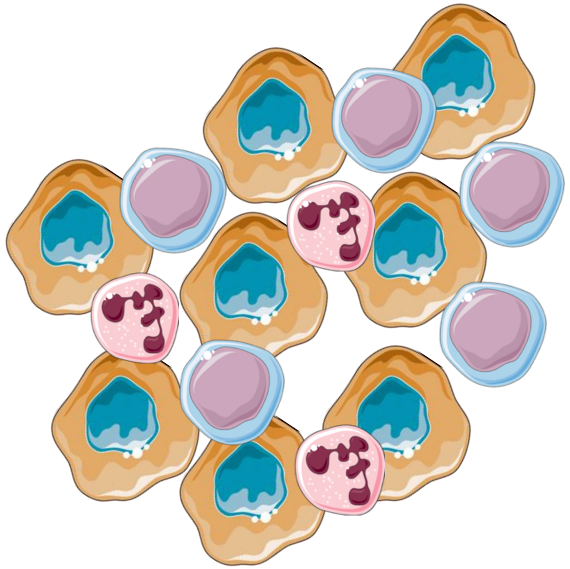
doi: https://doi.org/10.1101/2024.04.15.589472

》 In contrast to standard approaches to identifying cell types, **cellular niches** are defined by **spatially-dependent labels** obtained by combining **similarity in gene expression profiles** together with **spatial neighborhood structures**, **histological information** or **global structures in the tissue** Cellular niches typically describe a local structure in a tissue, for example, a **tumor niche or immune niche**, which vary both in terms of gene expression, but also in terms of the cell type compositions. Similarly to niches, tissue regions can be defined as spatially consistent cellular states across a larger-than-a-niche tissue context. Importantly tissue regions are commonly composed of multiple different niches and therefore describe higher-level structures in a tissue.

Let’s call them **spatially-aware clusters (SACs)**

Spatial clustering principles

Cluster cells based on the gene expression of:



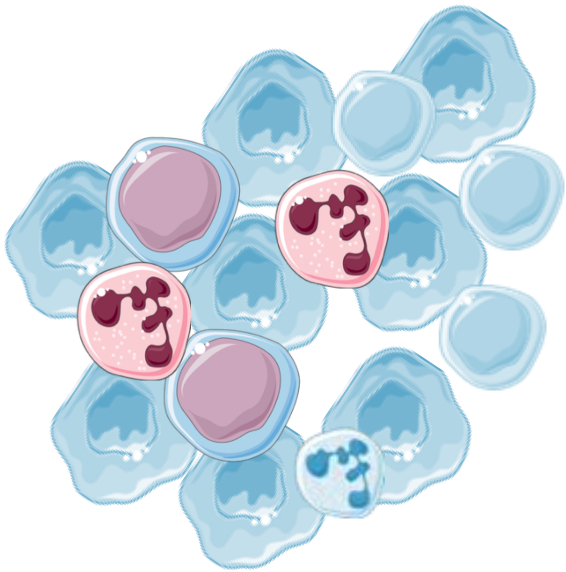
Spatial clustering principles

Cluster cells based on the gene expression of:

Cell

Spatial clustering principles

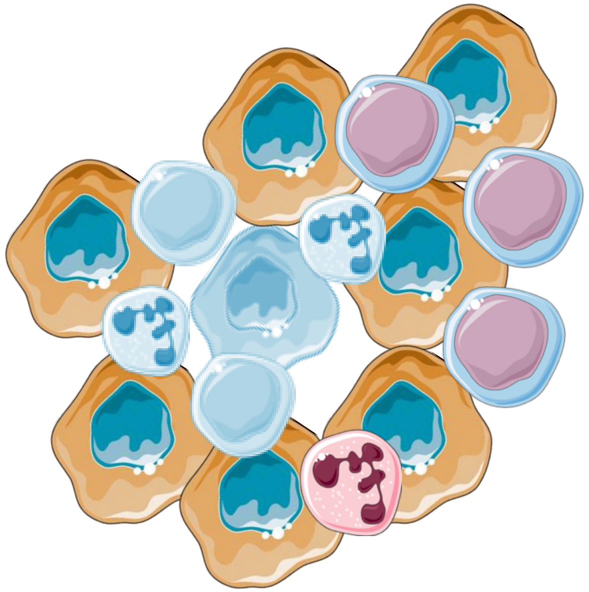
Cluster cells based on the gene expression of:



Cell
+
Immediate neighbors (niche)

Spatial clustering principles

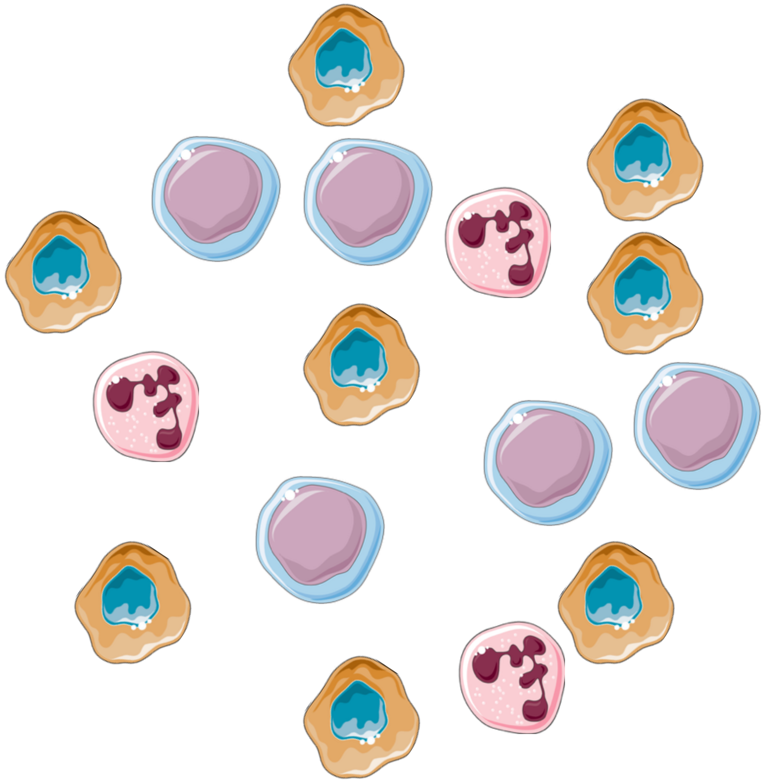
Cluster cells based on the gene expression of:



Cell
+
Immediate neighbors (niche)
+
Farther neighbors

Spatial domain identification approaches

Approach 1: proportion of cell types



Goltsev et al. "Deep profiling of mouse splenic architecture with CODEX multiplexed imaging." *Cell* (2018).
Schürch et al. "Coordinated cellular neighborhoods orchestrate antitumoral immunity at the colorectal cancer invasive front." *Cell* (2020).

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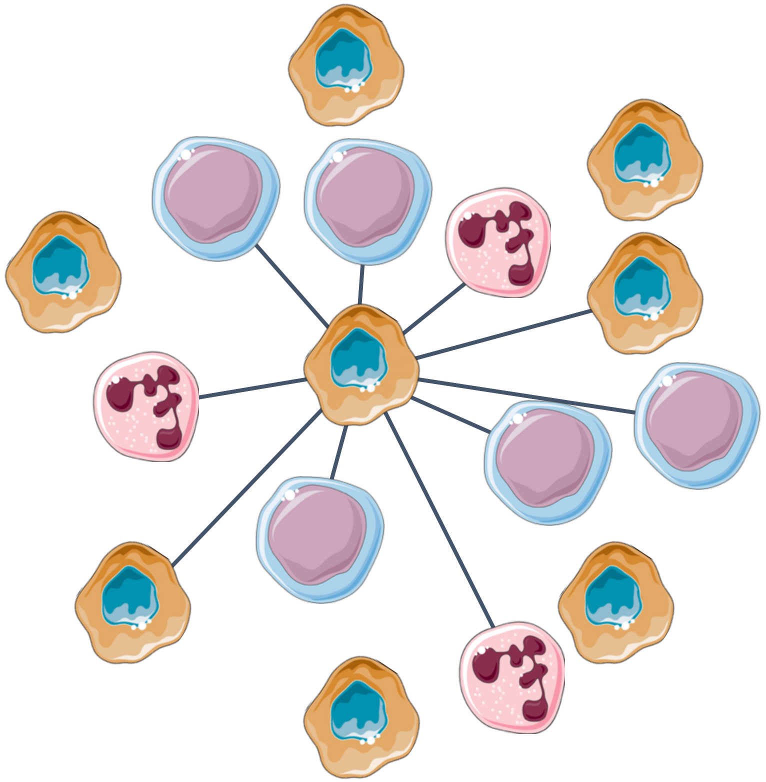

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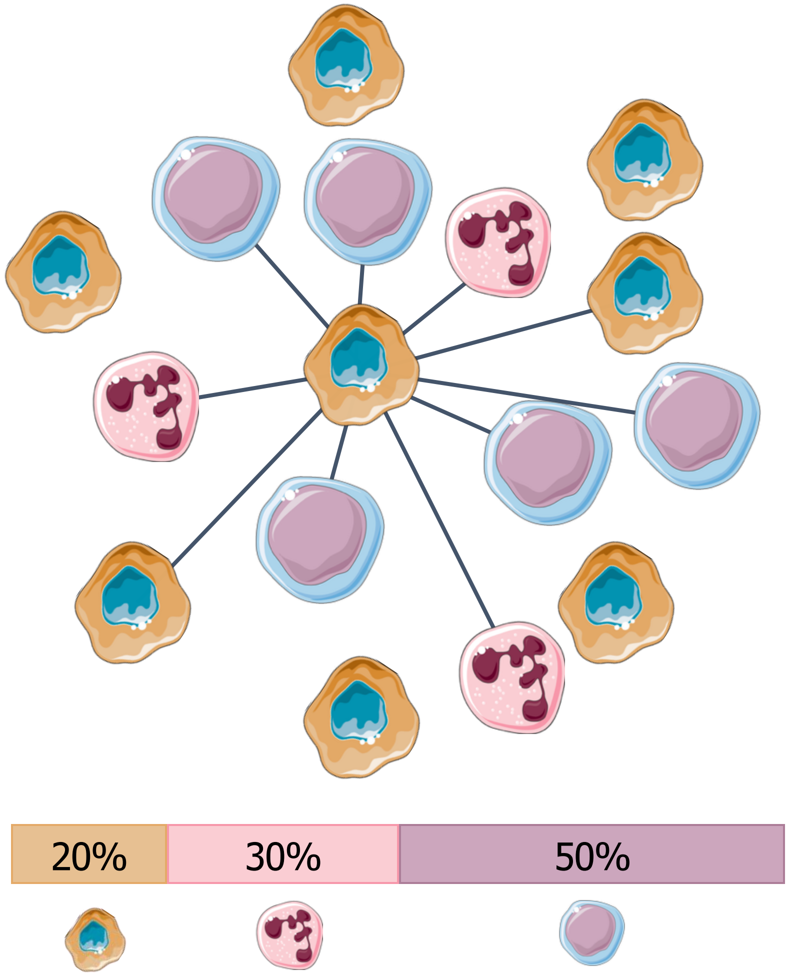

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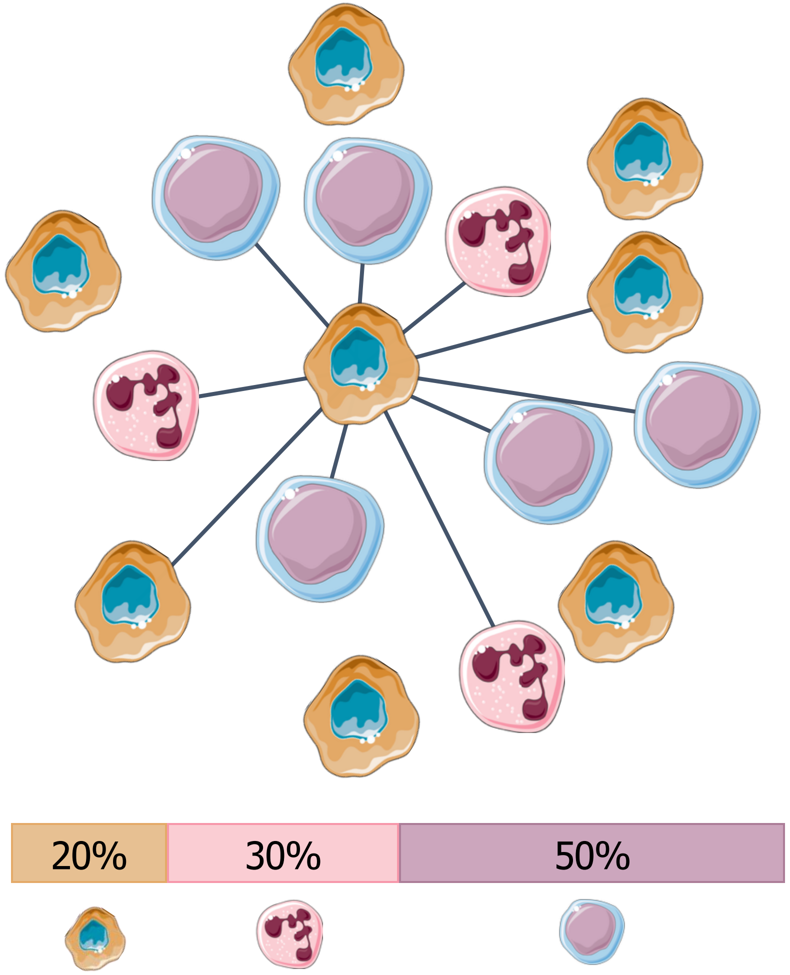
Approach 1: proportion of cell types

Steps

1. Compute neighbors proportion
   for every cell

2. Cluster cells
   based on their proportions



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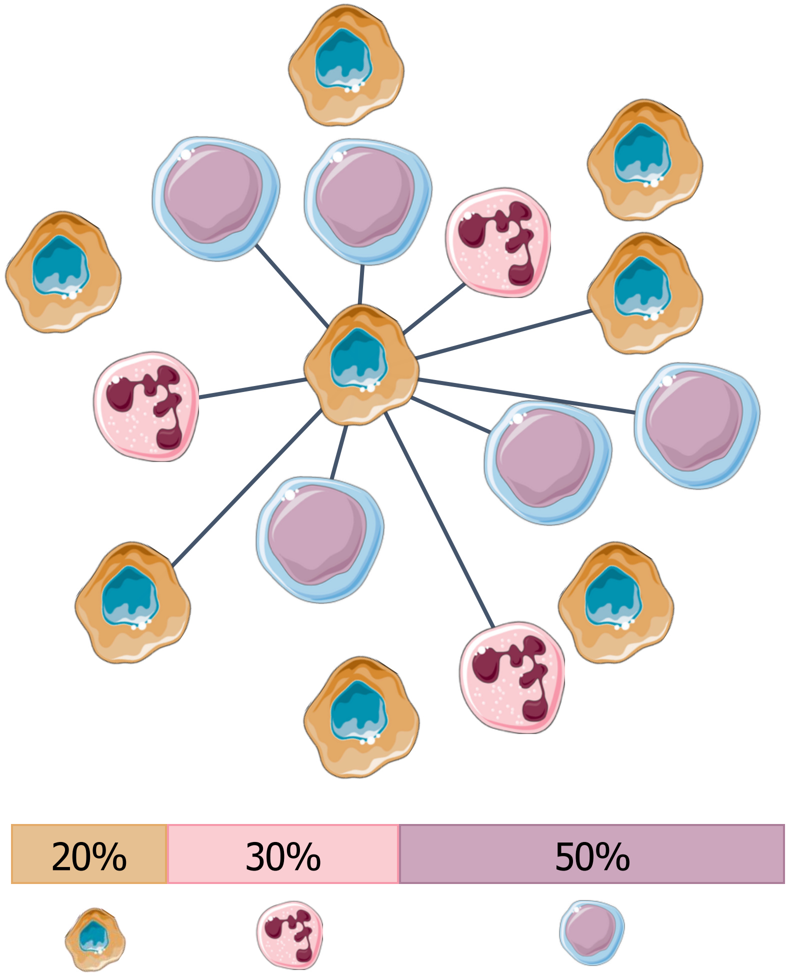

- Scalable
- Depends on manual annotations

- *How detailed should the annotation be?*

*- How to capture **variability** within cell types?*

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Schürch et al. "Coordinated cellular neighborhoods orchestrate antitumoral immunity at the colorectal cancer invasive front." *Cell* (2020).

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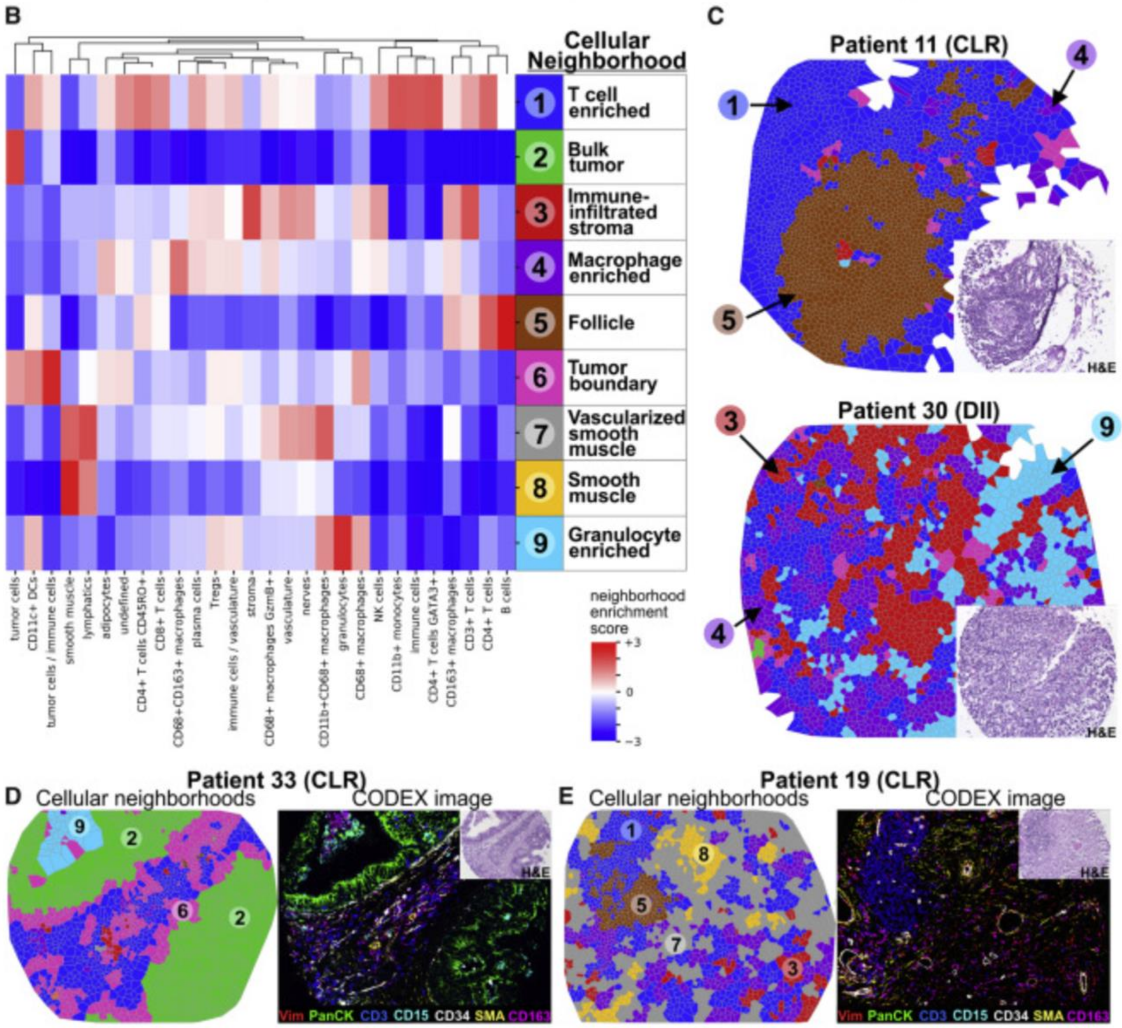

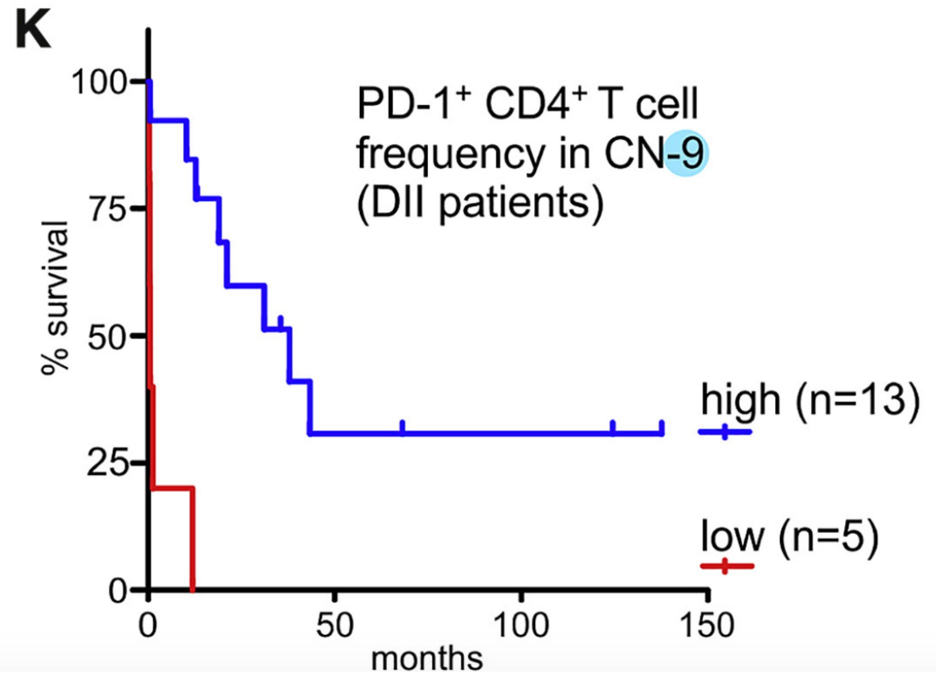

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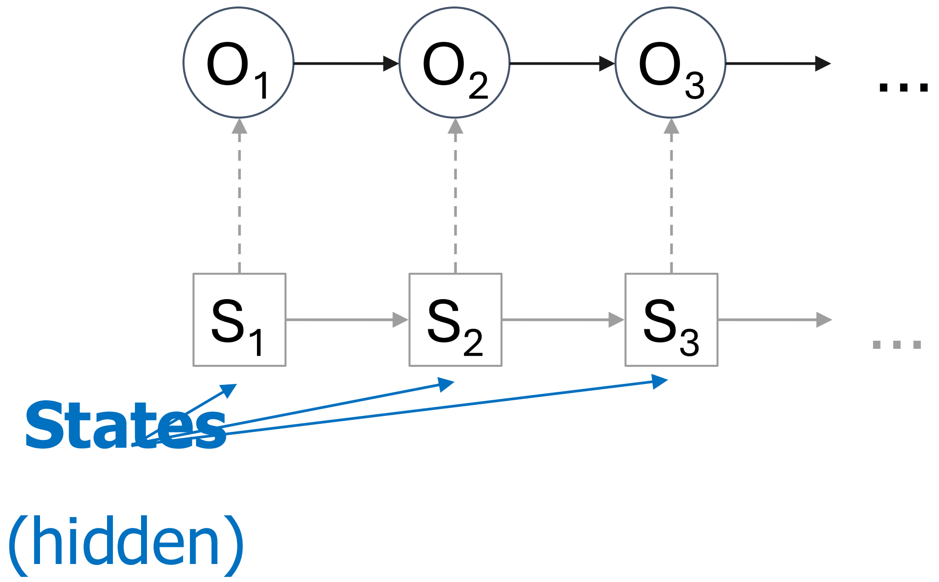
Approach 2: Hidden Markov Random Fields

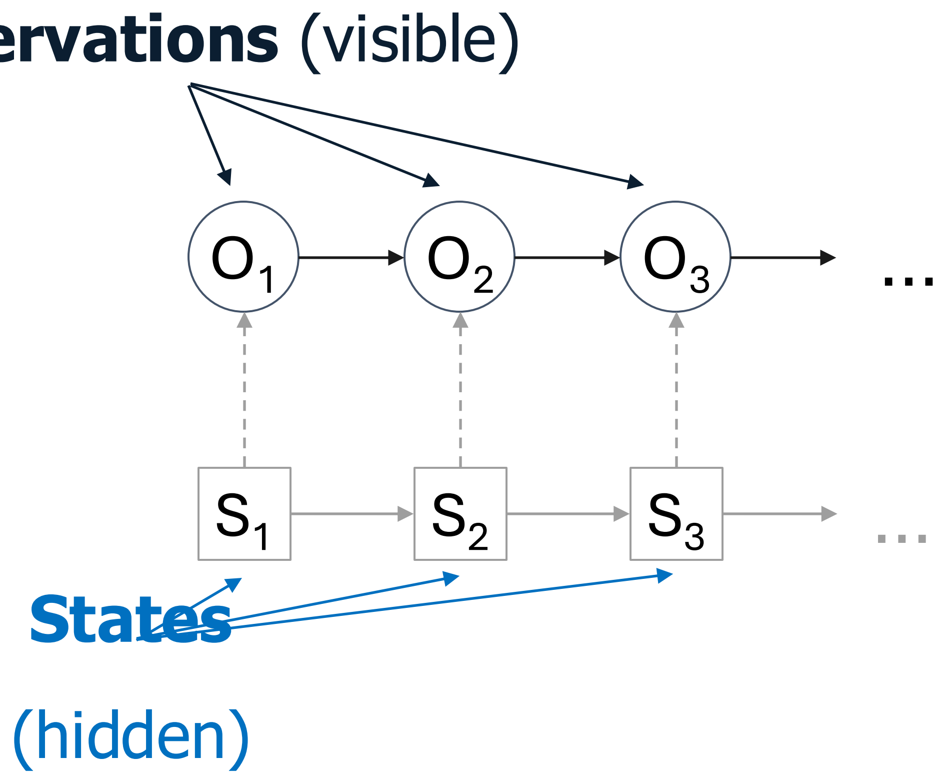
Hidden Markov Model



Sequences → 1D graphs

Approach 2: Hidden Markov Random Fields

Hidden Markov Model



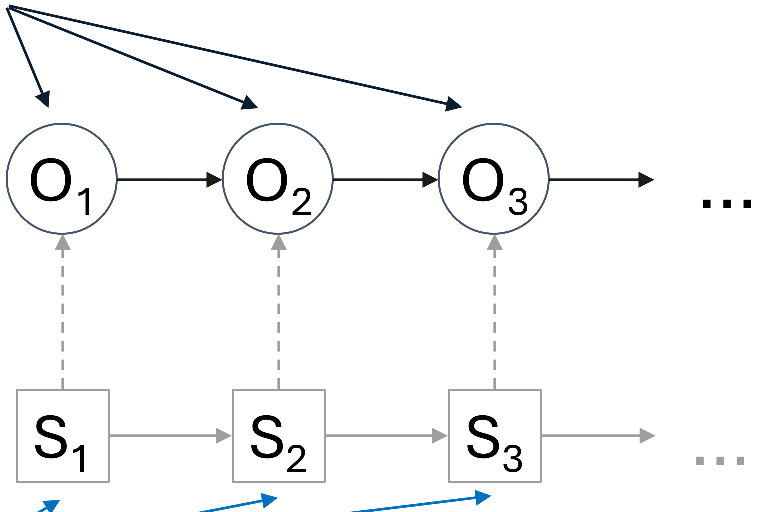
Sequences → 1D graphs

Approach 2: Hidden Markov Random Fields

Histone marks levels (high vs low)

Hidden Markov Model

Observations (visible)



Chromatin state (active vs inactive)

States (hidden)

Genome sequence

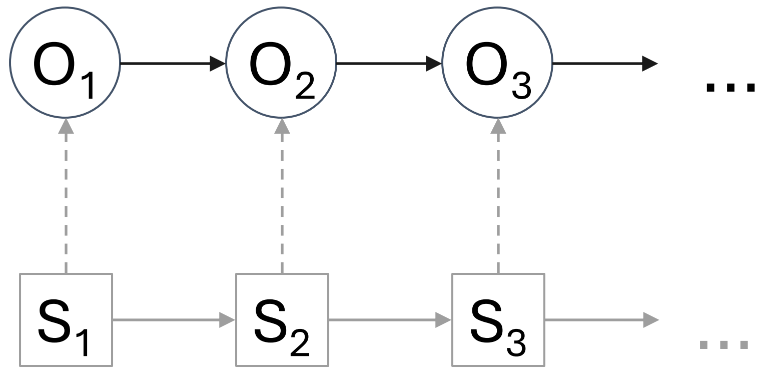
Sequences → 1D graphs

Approach 2: Hidden Markov Random Fields

Histone marks levels (high vs low)

Hidden Markov Model

Observations (visible)



Neighboring observations (O) are likely to have **similar states (S)**

Chromatin state (active vs inactive)

States (hidden)

Genome sequence

Sequences → 1D graphs

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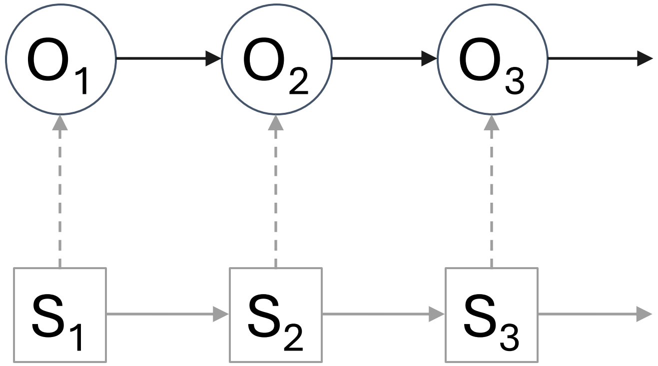
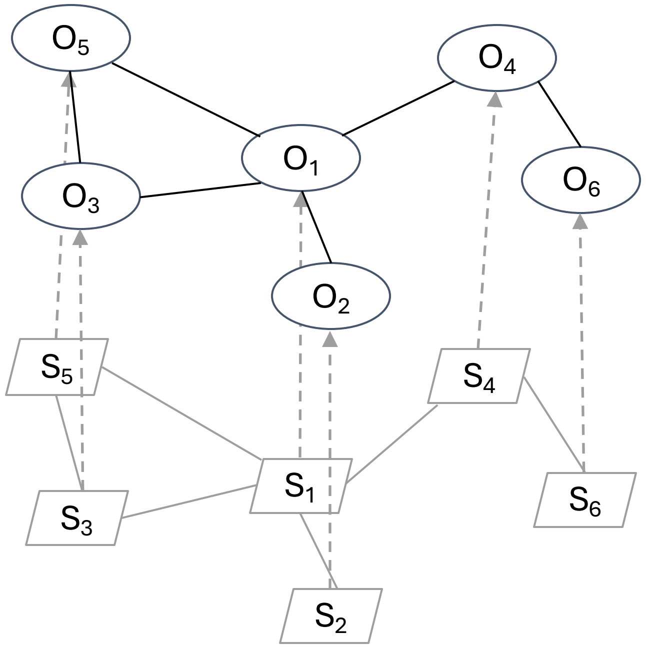


Hidden Markov Random Field



Sequences → 1D graphs

Arbitrary graphs

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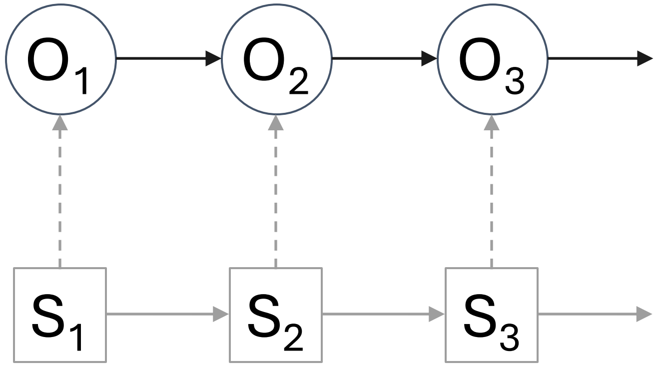
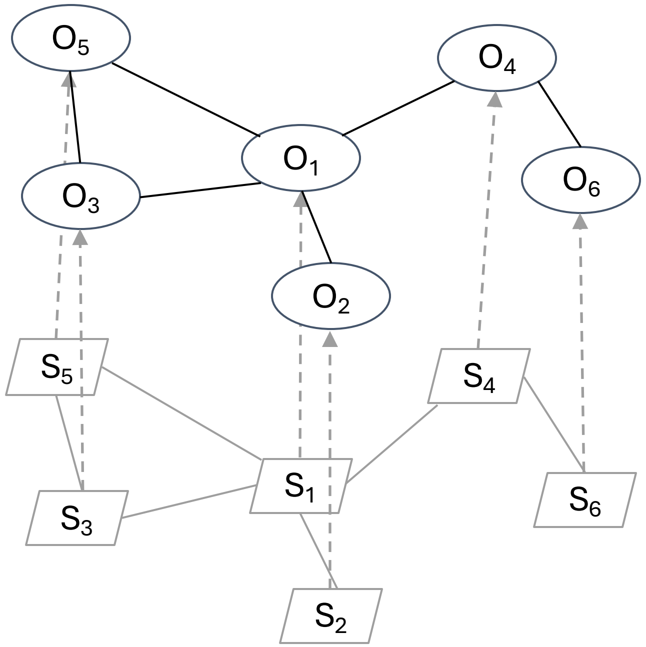


Sequences → 1D graphs

Hidden Markov Random Field

Gene expression



Spatial domain

Arbitrary graphs

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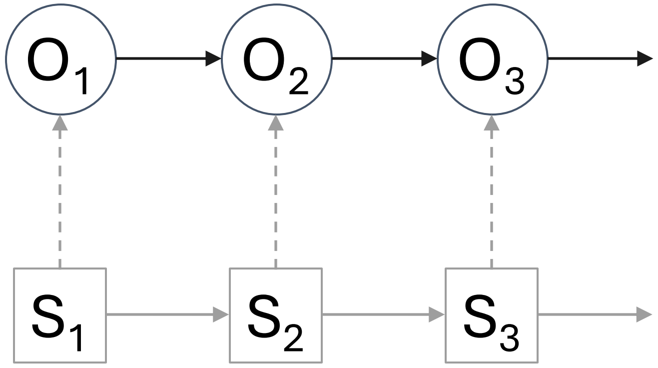

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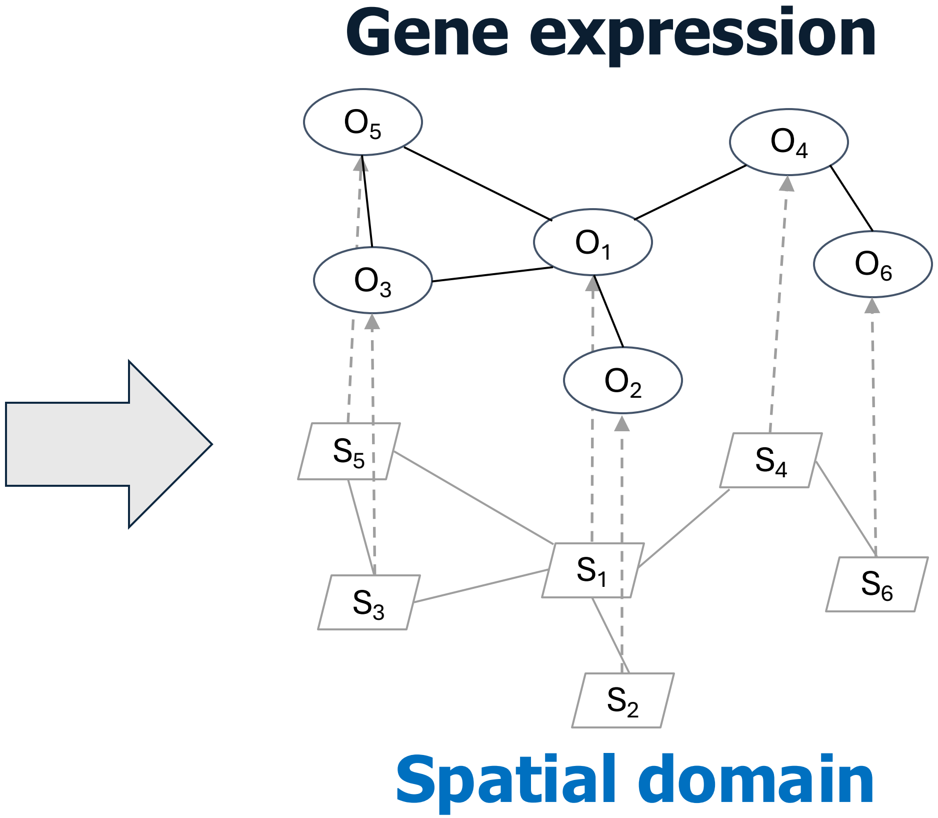

Arbitrary graphs

- Medium scalability
- No annotation required
  Works directly on gene expression

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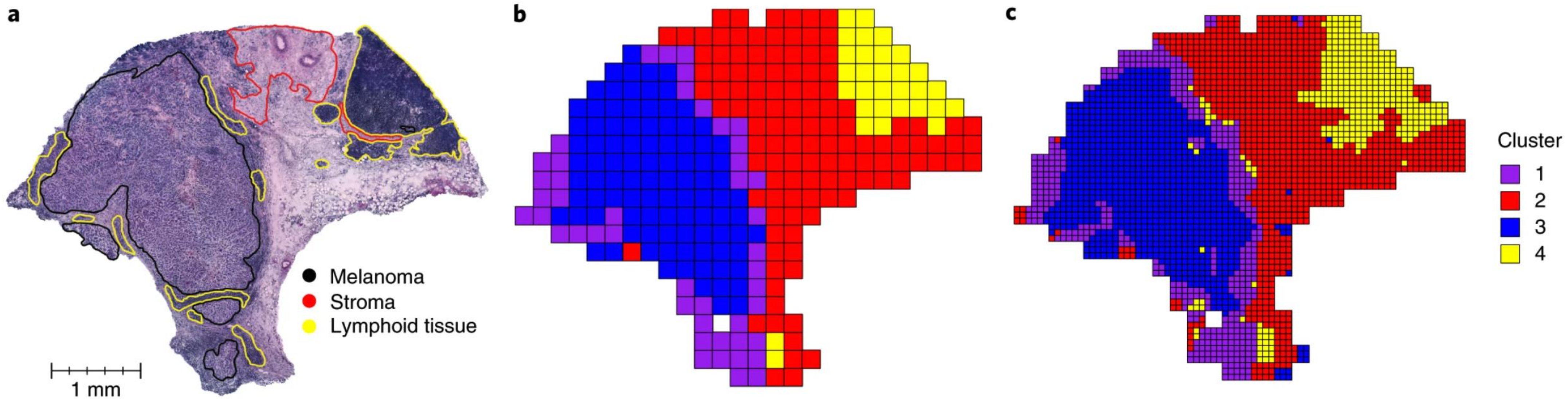

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Approach 3: Graph Neural Networks

Nodes with **similar neighborhoods** have **similar representations** (i.e. vectors)

Hu et al. "SpaGCN: Integrating gene expression, spatial location and histology to identify spatial domains and spatially variable genes by graph convolutional network." *Nature methods* (2021).
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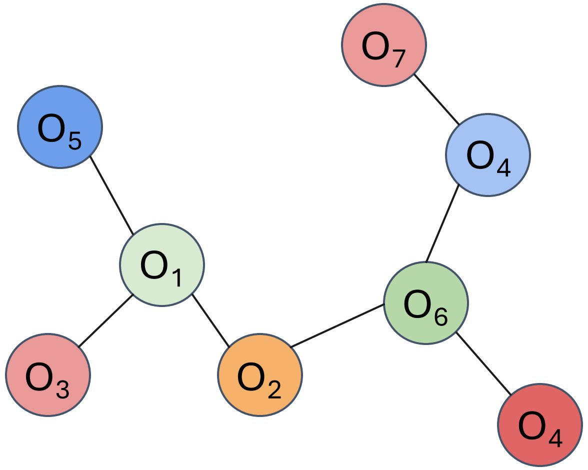

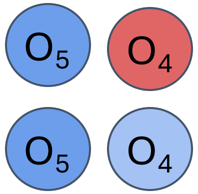

Different cell types → very different gene expression

Variability within cell type → slightly different gene expression

A Gentle Introduction to Graph Neural Networks
https://distill.pub/2021/gnn-intro/

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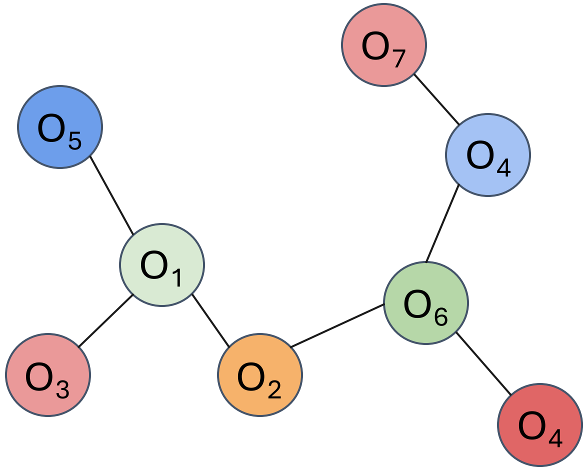

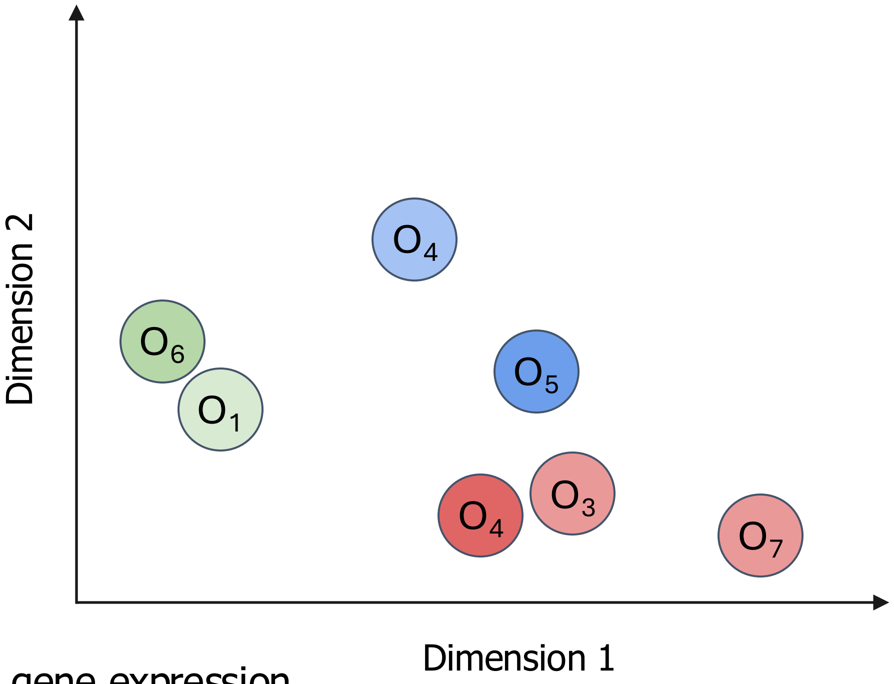

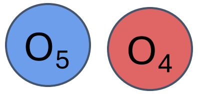

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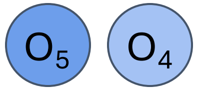

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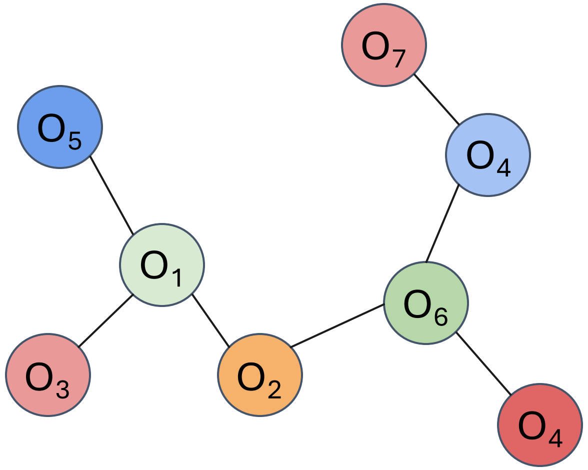
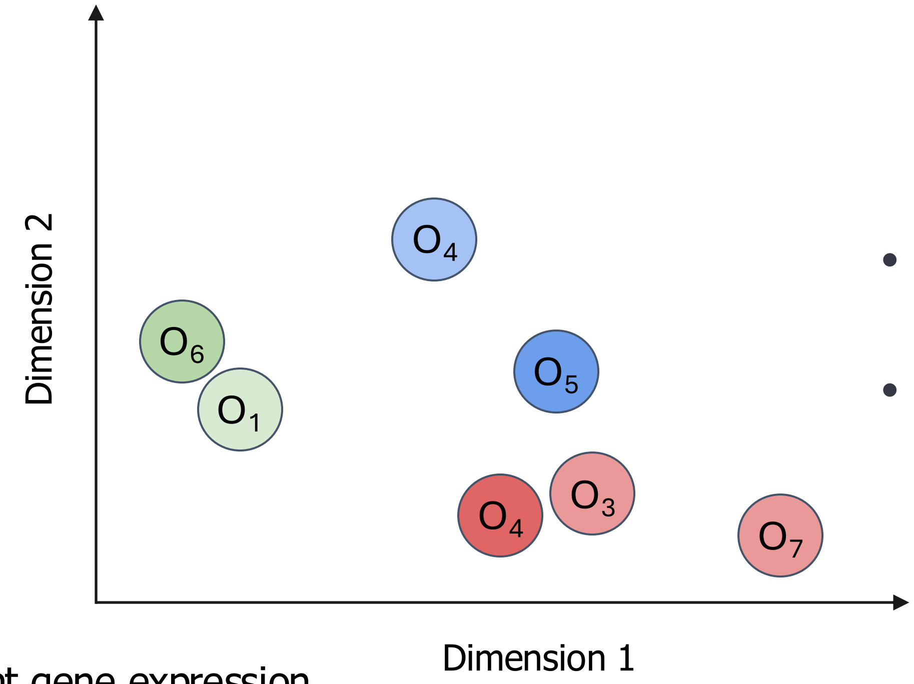
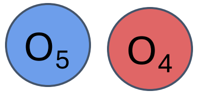




- Low/medium scalability
- No annotation required
  Works directly on gene expression



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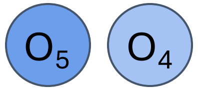

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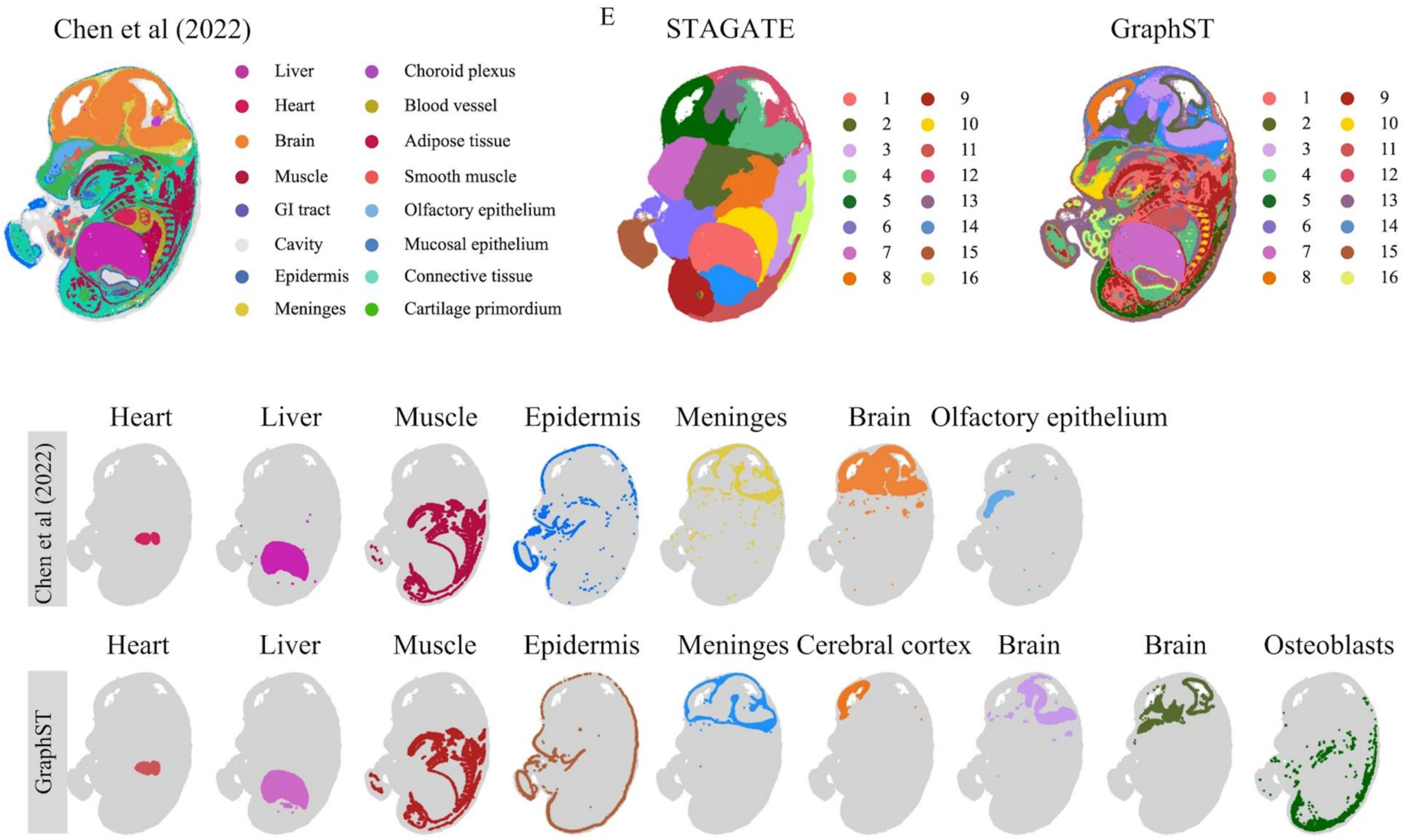

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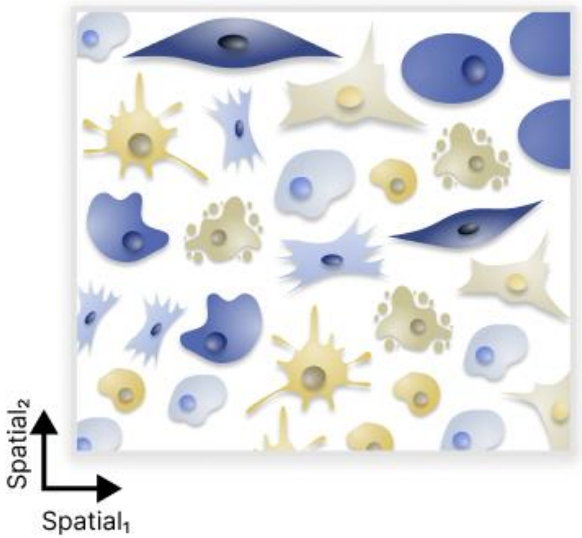
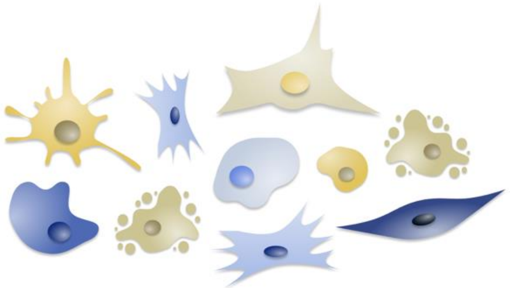
Defining spatial graphs of cells based on spatial proximity

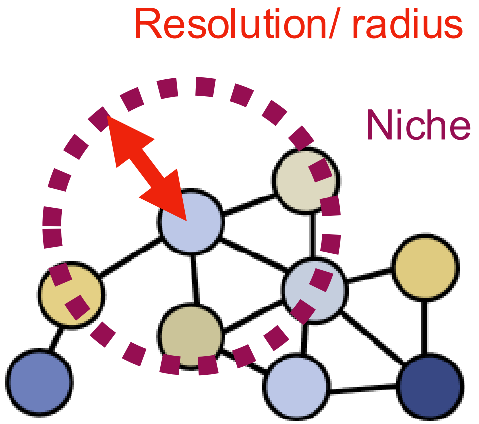


Segmented targeted spatial omics

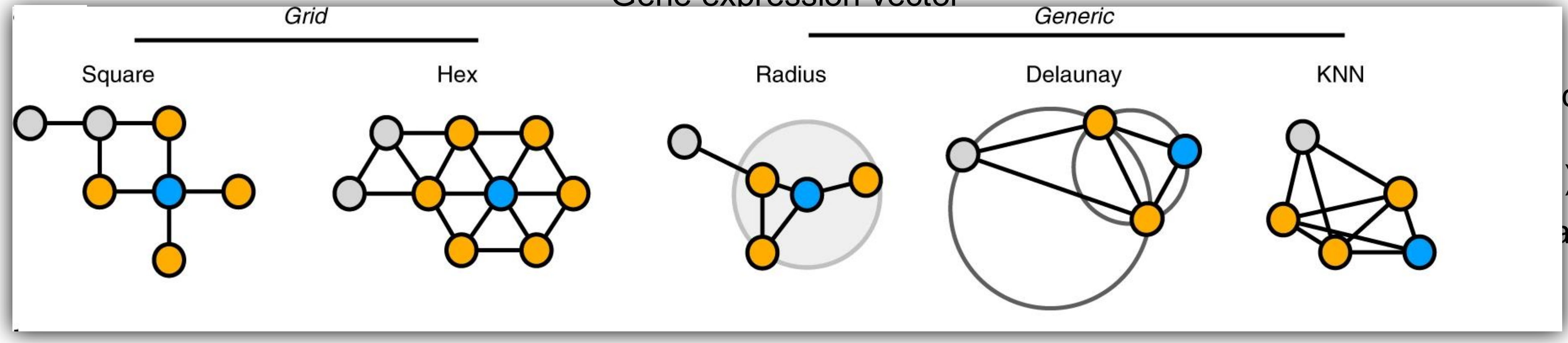


Cells resolved at their physical location

- Coordinates
- Gene expression vector



Spatial graph of cells



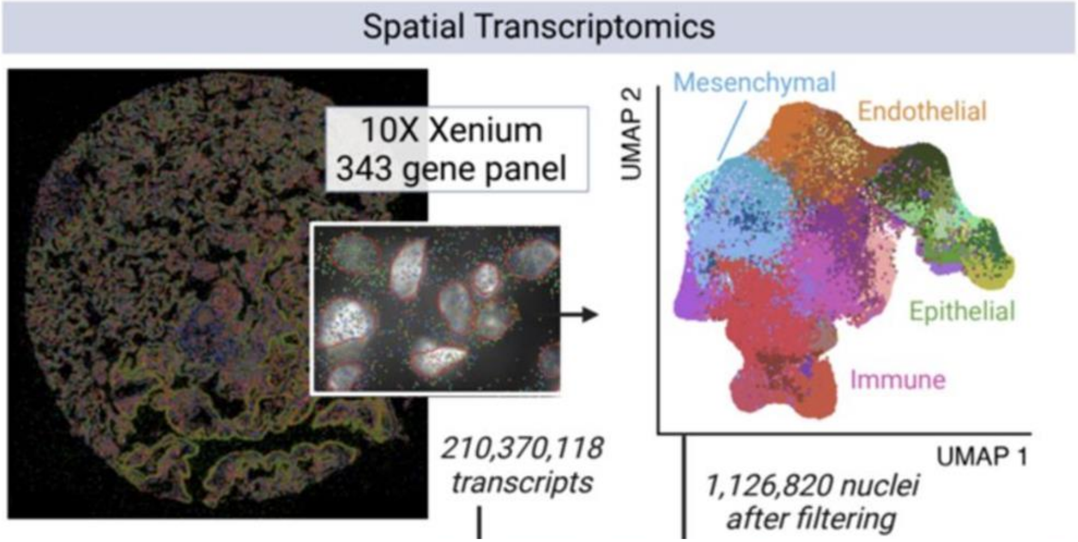
cency matrix

) ≤ r with d being

ance between two

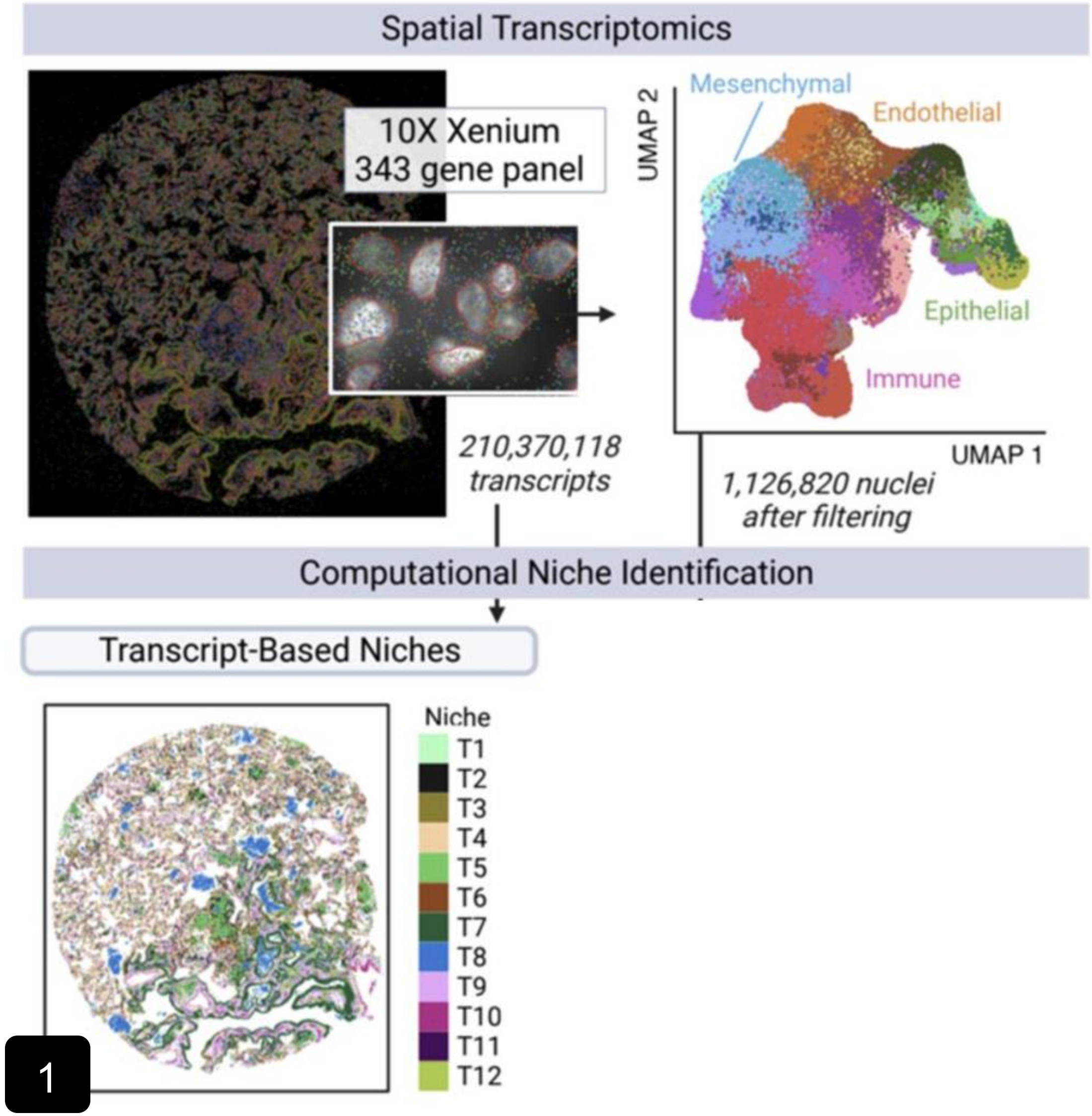
Palla, G., Spitzer, H., Klein, M. *et al.* Squidpy: a scalable framework for spatial omics analysis. *Nat Methods* **19**, 171–178 (2022). https://doi.org/10.1038/s41592-021-01358-2

How to define niche labels in practice?



Vannan, A., *et al.* Image-based spatial transcriptomics identifies molecular niche dysregulation associated with distal lung remodeling in pulmonary fibrosis. b*ioRix*, (2023).

How to define niche labels in practice?



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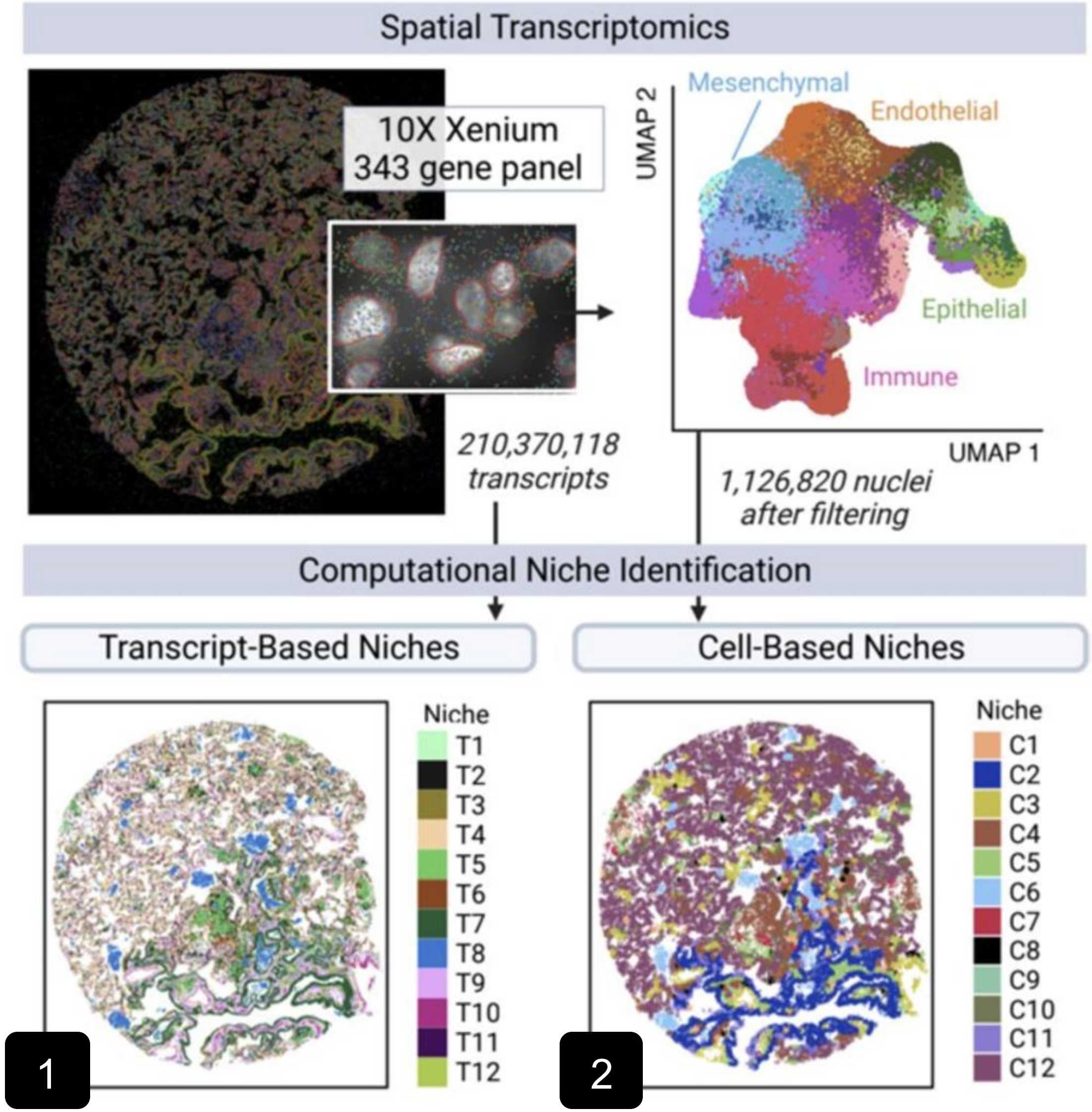

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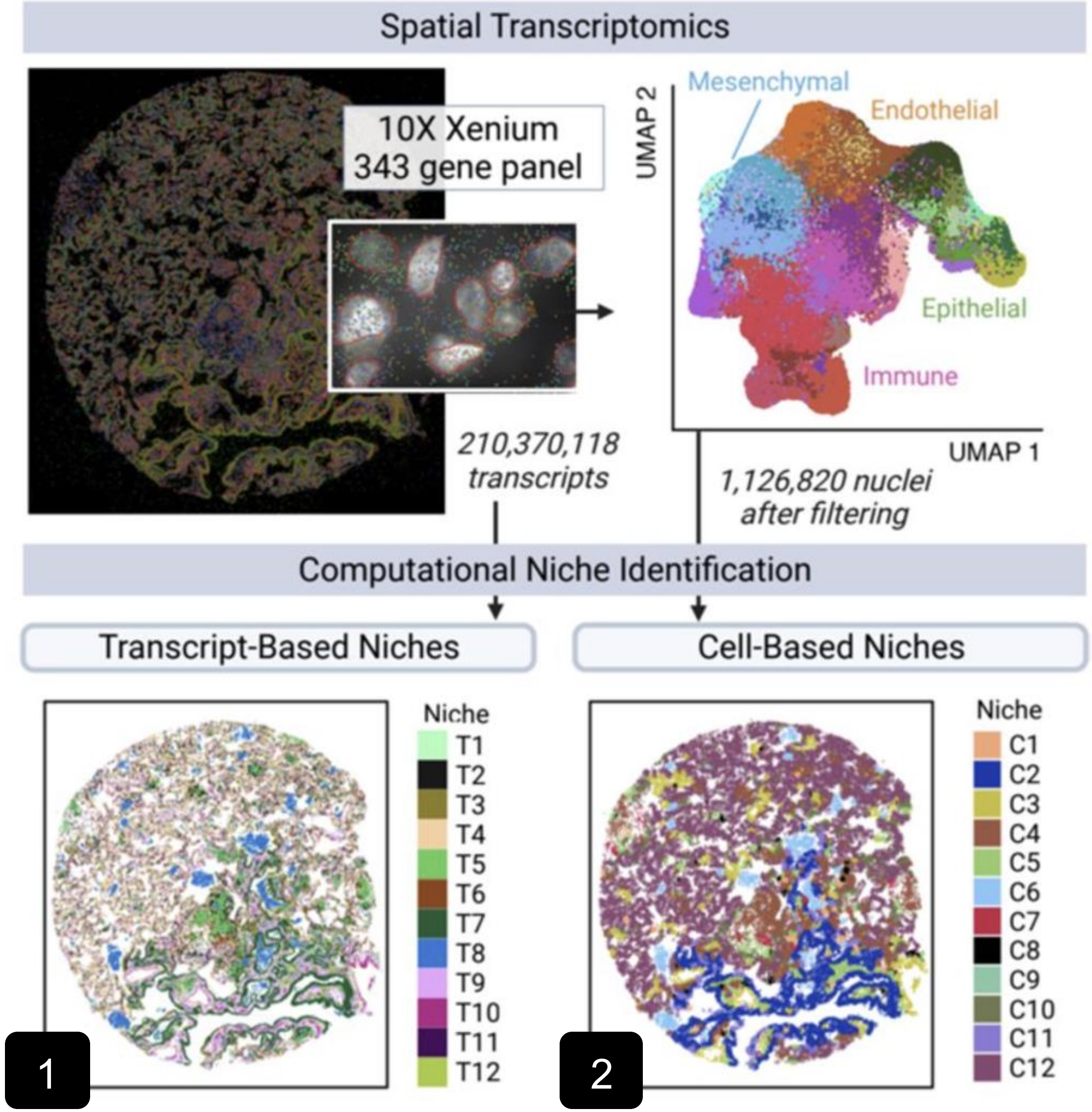
How to define niche labels in practice?

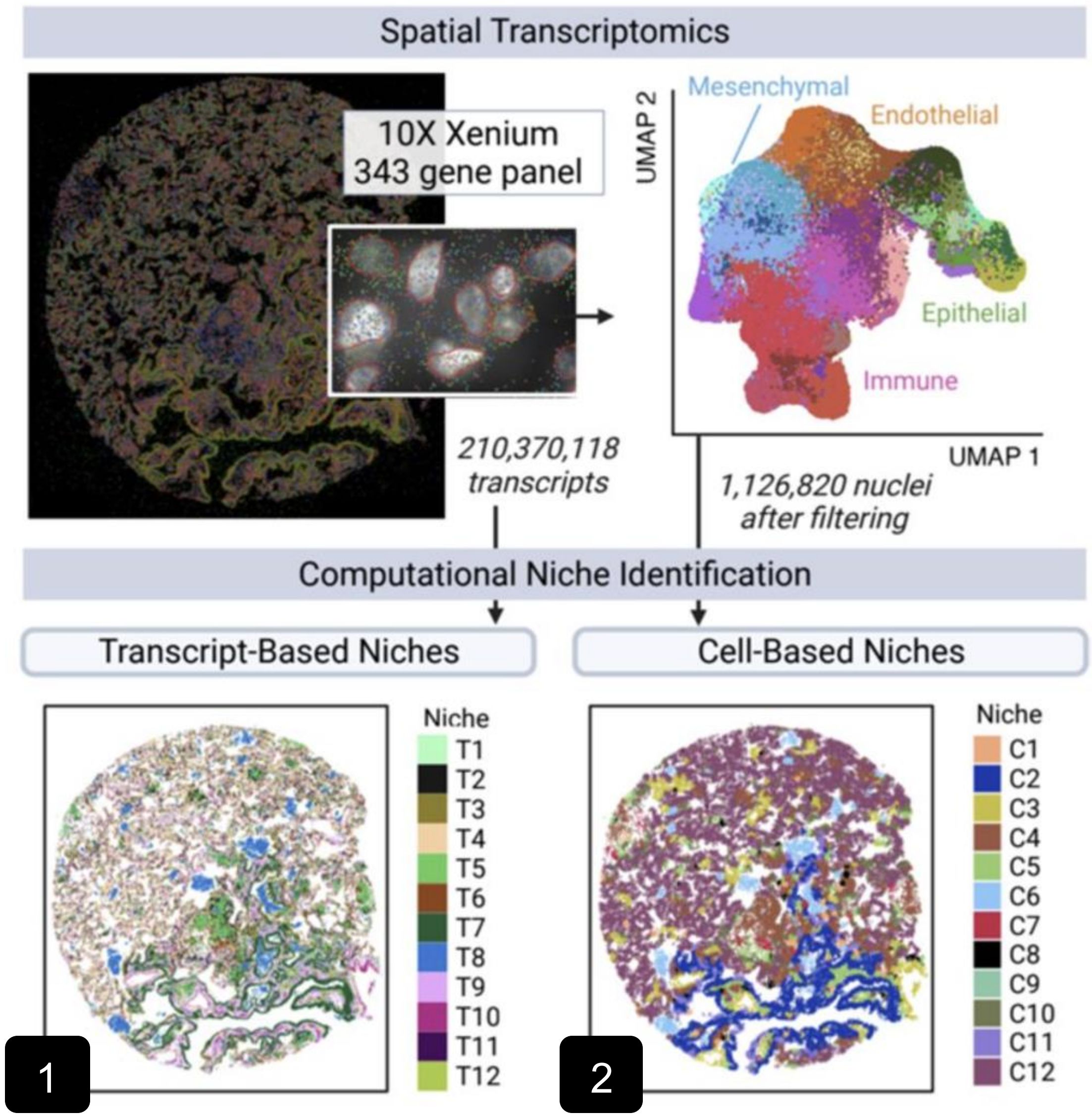


3

Communication-based Niches

Vannan, A., *et al.* Image-based spatial transcriptomics identifies molecular niche dysregulation associated with distal lung remodeling in pulmonary fibrosis. *bioRix*, (2023).

How to define niche labels in practice?



3

Communication-based Niches

Note:

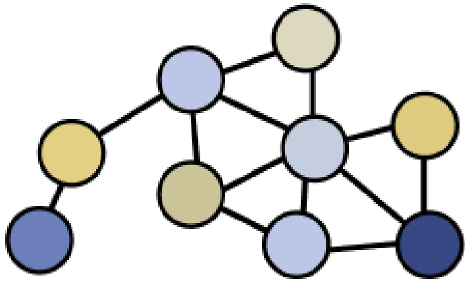
- Niche labels can be very tissue dependent
- Niche labels strongly depend on the specific use-case you are interested in
- Batch-effects might not be only technical variability, but biological differences

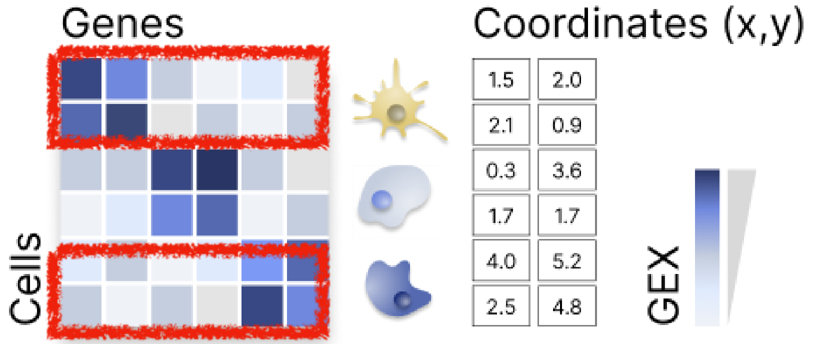
Vannan, A., *et al.* Image-based spatial transcriptomics identifies molecular niche dysregulation associated with distal lung remodeling in pulmonary fibrosis. *bioRix*, (2023).

Transcription-based niches

i.e. define a niche based on the expression profile of its surrounding neighbours

Spatial graph of cells





$A \in \{0;1\}^{\{c \times c\}}$ adjacency matrix

$Y \in \mathbb{R}^{+,\{c \times g\}}$ gene expression matrix

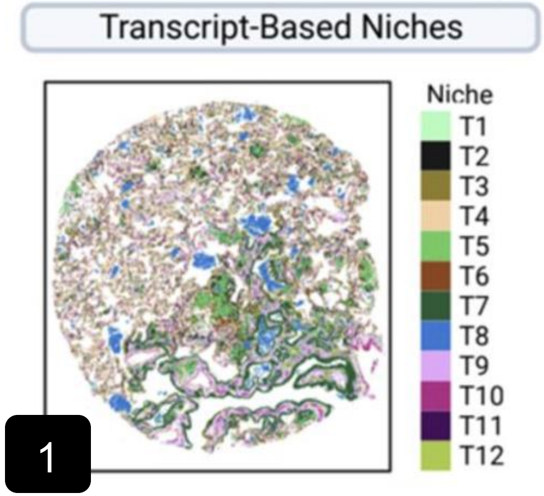
with c number of cells,

g number of genes

Aggregation function

$$f(Y, A)$$

Trivial example: pseudo Visium spots $\sum Y \cdot A$

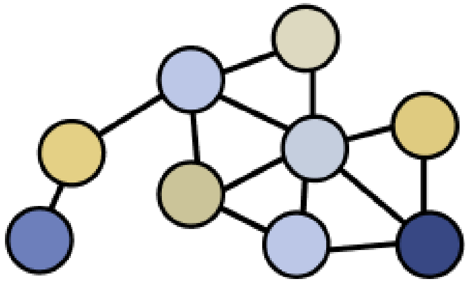


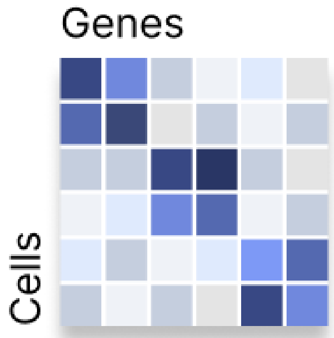
Vannan, A., *et al.* *bioRix*, (2023).

Transcription-based niches

i.e. define a niche based on the expression profile of its surrounding neighbours

Spatial graph of cells





Coordinates (x,y)

1.5 2.0

$A \in \{0;1\}^{\{c \times c\}}$ adjacency matrix

sion matrix

es

Article | Open access | Published: 27 February 2024

BANKSY unifies cell typing and tissue domain segmentation for scalable spatial omics data analysis

Vipul Singhal, Nigel Chou, Joseph Lee, Yifei Yue, Jinyue Liu, Wan Kee Chock, Li Lin, Yun-Ching Chang, Erica Mei Ling Teo, Jonathan Aow, Hwee Kuan Lee, Kok Hao Chen & Shyam Prabhakar

Nature Genetics **56**, 431–441 (2024) | Cite this article

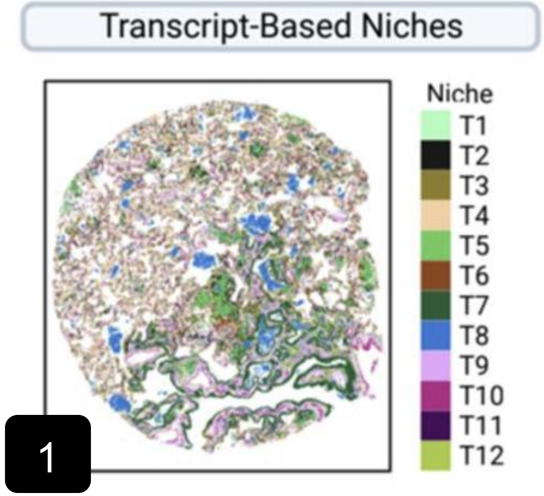
Aggreg

For each gene q, we computed the mean expression in the neighborhood of cell $u \in \mathcal{U}$ as:

$$M_u^{(q)} = \sum_{v \in \eta^u_{k_{\text{geom}}}} g_v^{(q)} \Gamma_{uv}^{k_{\text{geom}}}$$

where $g_v^{(q)}$ is the expression of the qth gene in neighboring cell v and the term $\Gamma_{uv}^{k_{\text{geom}}}$ is a Gaussian weighting envelope defined as:

$$\Gamma_{uv}^{k_{\text{geom}}} = \frac{\exp \frac{-r_{uv}^2}{r_{u\bar{v}}^2}}{\sum_{w \in \eta^u_{k_{\text{geom}}}} \left(\exp \frac{-r_{uw}^2}{r_{u\bar{v}}^2} \right)}$$

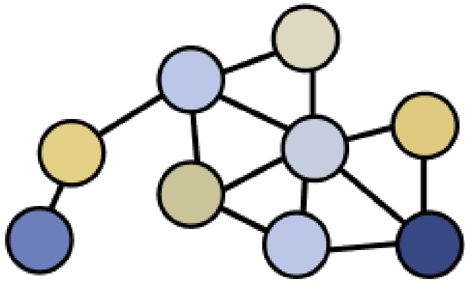


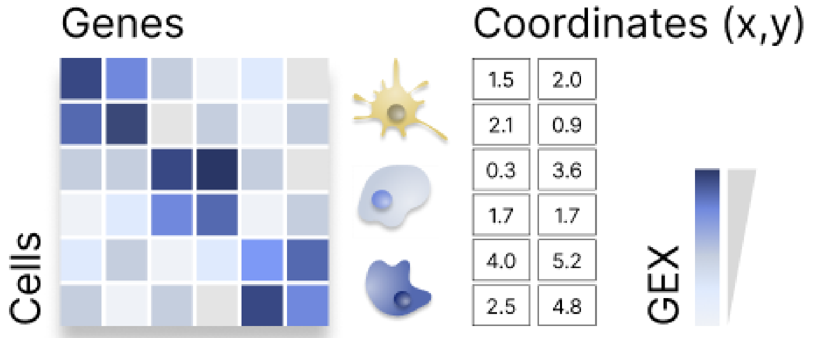
Vannan, A., *et al.* bioRix, (2023).

Transcription-based niches

i.e. define a niche based on the expression profile of its surrounding neighbours

Spatial graph of cells





$A \in \{0;1\}^{\{c \times c\}}$ adjacency matrix

$Y \in \mathbb{R}^{+,\{c \times g\}}$ gene expression matrix

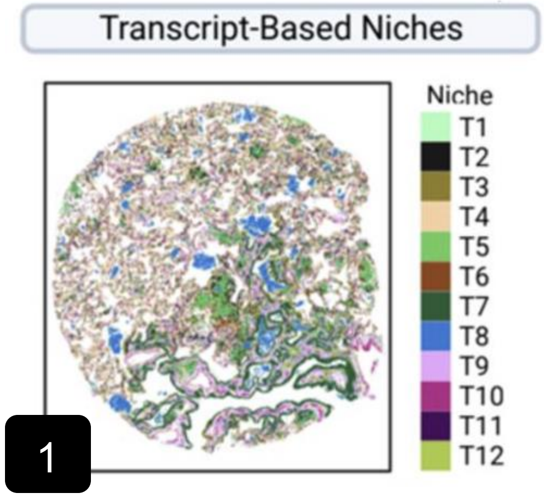
with c number of cells,

g number of genes

Aggregation function

$f(Y, A)$

Trivial example: pseudo Visium $\sum Y \cdot A$




```python
import scanpy as sc

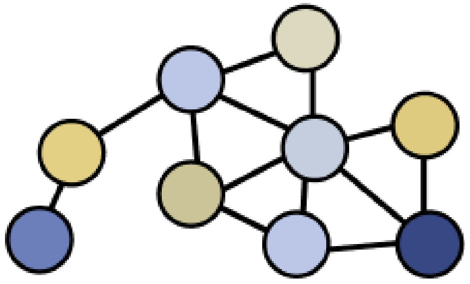
sc.pp.neighbors(adata, use_rep="X_niche",
 key_added="niche_connectivities")
sc.tl.leiden(adata,
 adjacency="niche_connectivities")
```

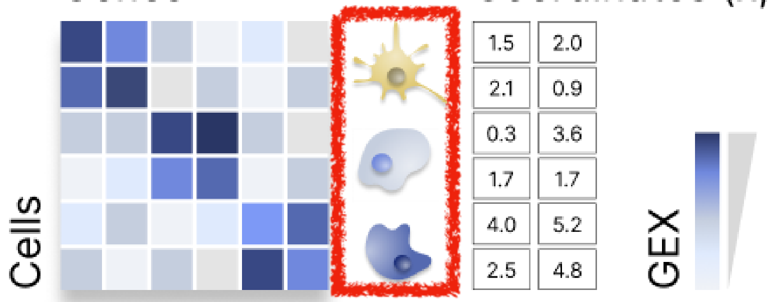
Vannan, A., *et al.* *bioRix*, (2023).

Cell-based niches

i.e. define a niche based on the composition of neighbouring cell types

Spatial graph of cells





$A \in \{0;1\}^{\{c \times c\}}$ adjacency matrix

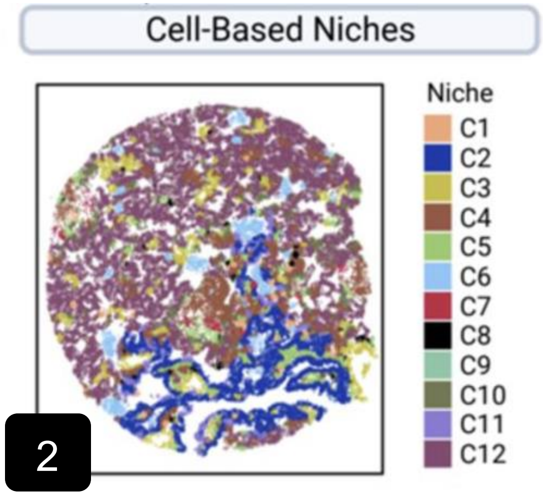
$X \in \{0,1\}^{\{c \times d\}}$ one-hot encoded cell-type matrix

with c number of cells,

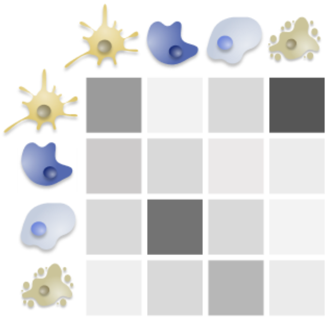
d number of cell-types

Aggregation function

$f(X, A)$



Cell-type composition of the niche



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import scanpy as sc

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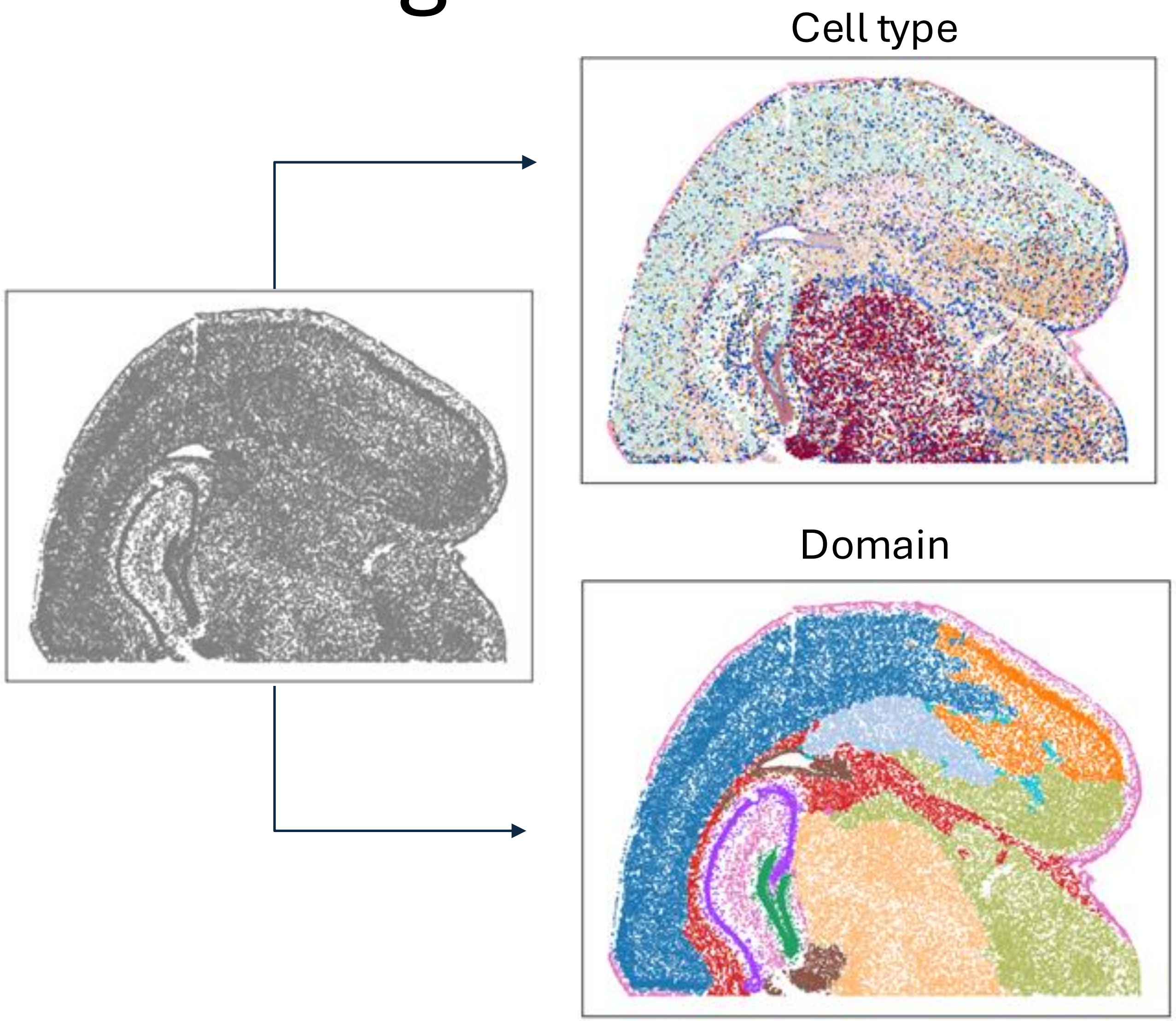
sc.tl.leiden(adata,
 adjacency="niche_connectivities")
```

Vannan, A., *et al.* bioRix, (2023).

Many different methods for spatially aware clustering



Maple
BANKSY
PRECAST
GraphST
SPICEMIX
CellCharter
STAGATE
scMEB
SpaceFlow
DR-SC
SCAN-IT
SOTIP
Stardust
...

Which method is the best? → benchmark

JOURNAL ARTICLE

Benchmarking cell-type clustering methods for spatially resolved transcriptomics data

Andrew Cheng, Guanyu Hu

Briefings in Bioinformatics
https://doi.org/10.1093/bi
Published: 21 November

Analysis | Published: 15 March 2024

Benchmarking spatial clustering methods with spatially resolved transcriptomics data

Zhiyuan Yuan, Fangyuan Zhao, Senlin Lin

Nature Methods **21**, 712–722 (2024) | Cite t

13k Accesses | **5** Citations | **45** Altmetric

Research | Open access | Published: 09 August 2024

Benchmarking clustering, alignment, and integration methods for spatial transcriptomics

Yunfei Hu, Manfei Xie, Yikang Li, Mingxing Rao, Wenjun Shen, Can Luo, Haoran Qin, Jihoon Baek & Xin Maizie Zhou

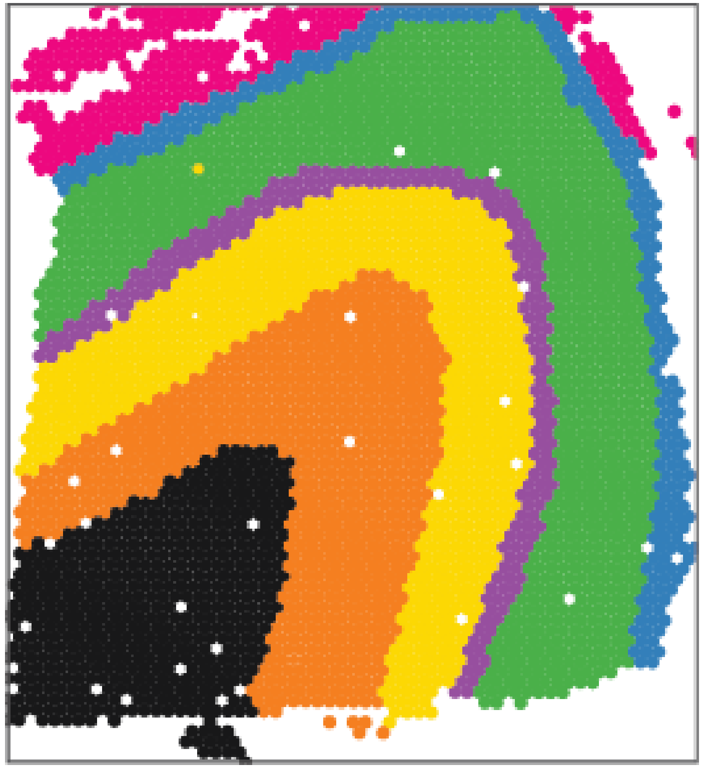
Genome Biology **25**, Article number: 212 (2024) | Cite this article

3804 Accesses | **2** Citations | **20** Altmetric | Metrics

Can we use these existing benchmarks in some way?

Spatially aware clustering methods

LIBD DLPFC
Human brain
12 samples



- Used by almost all methods
- Method papers compare to older methods

Method X
Method X on LIBD

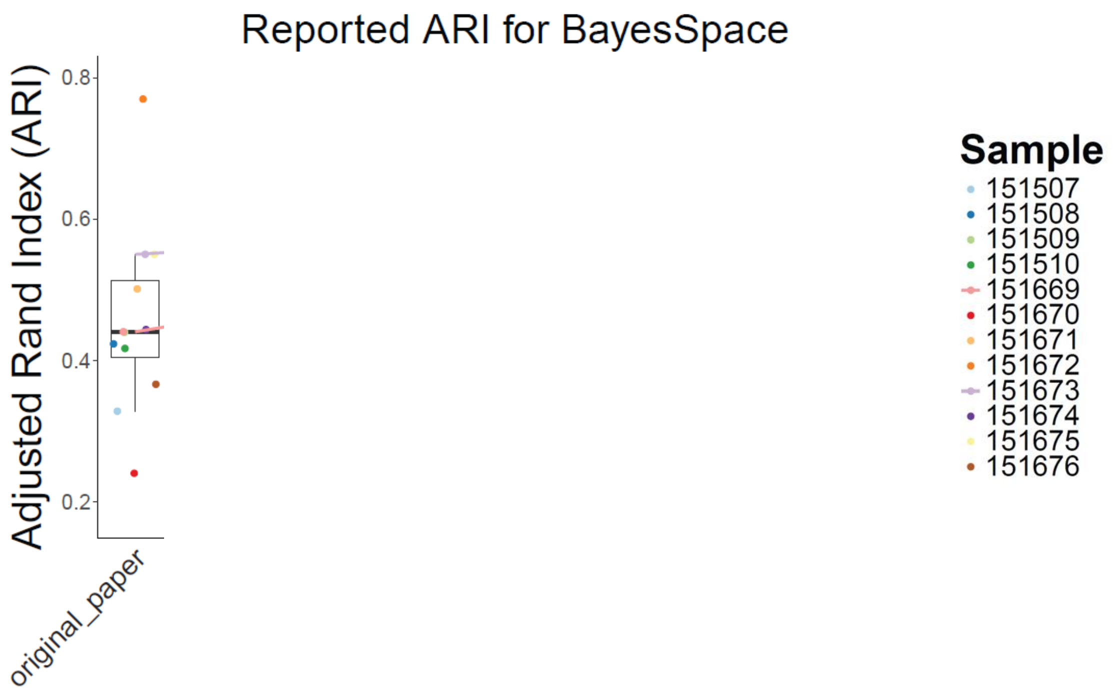
Method Y
Method Y on LIBD
Method X on LIBD

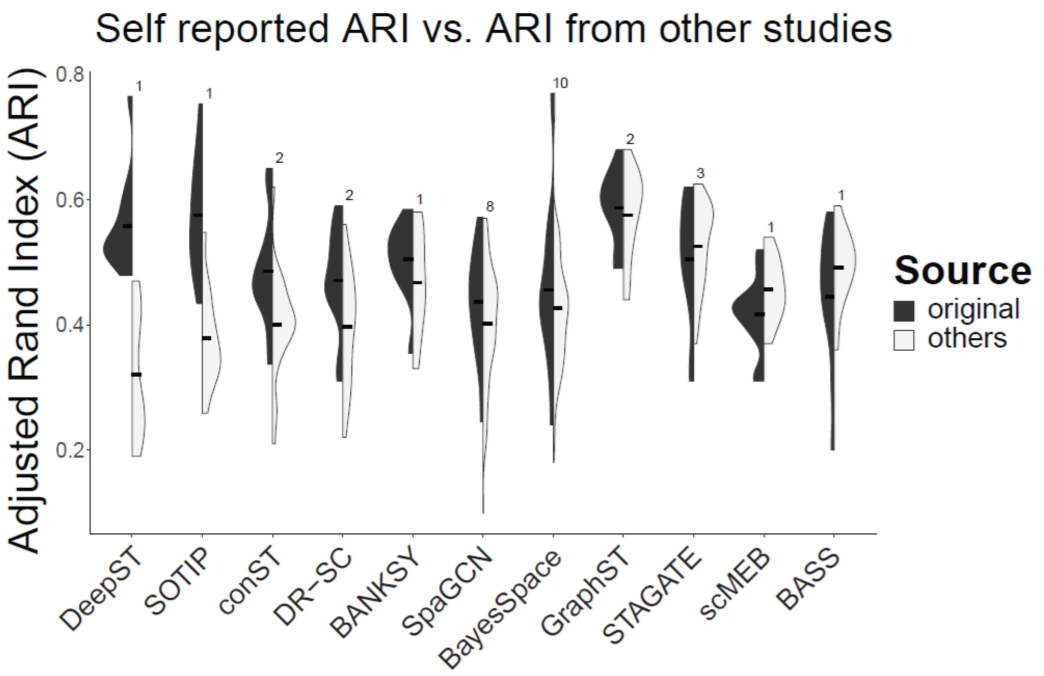
Method Z
Method Z on LIBD
Method Y on LIBD
Method X on LIBD

Benchmark
Method X on LIBD
Method Y on LIBD
Method Z on LIBD

These results should be identical… right?

Use existing benchmarks

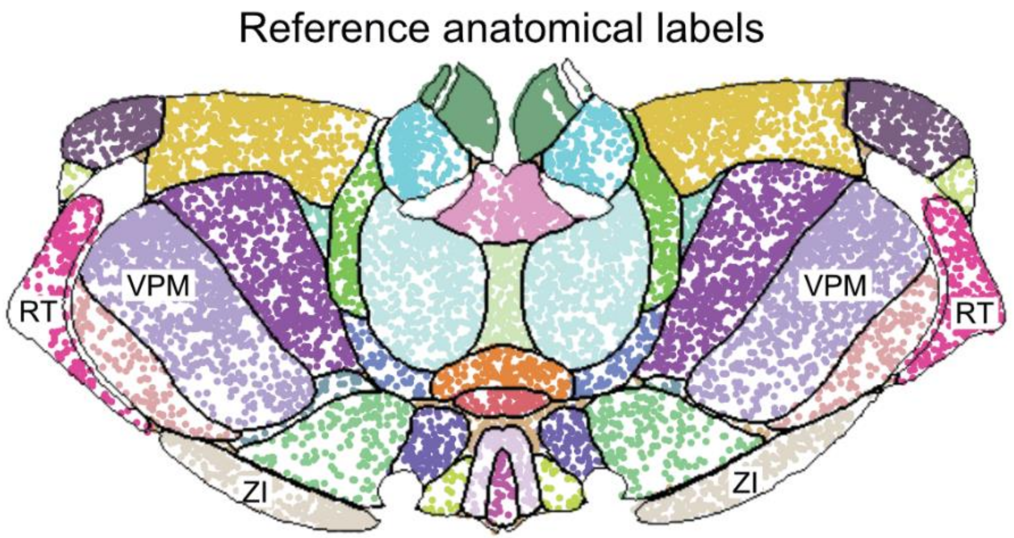




1. What causes this variability? → hyperparameters, preprocessing, ...
2. Are we evaluating the methods correctly?

Reliance on the ground truth

- Is the ground truth meant to rank these methods?



?

Maple
BANKSY
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scMEB
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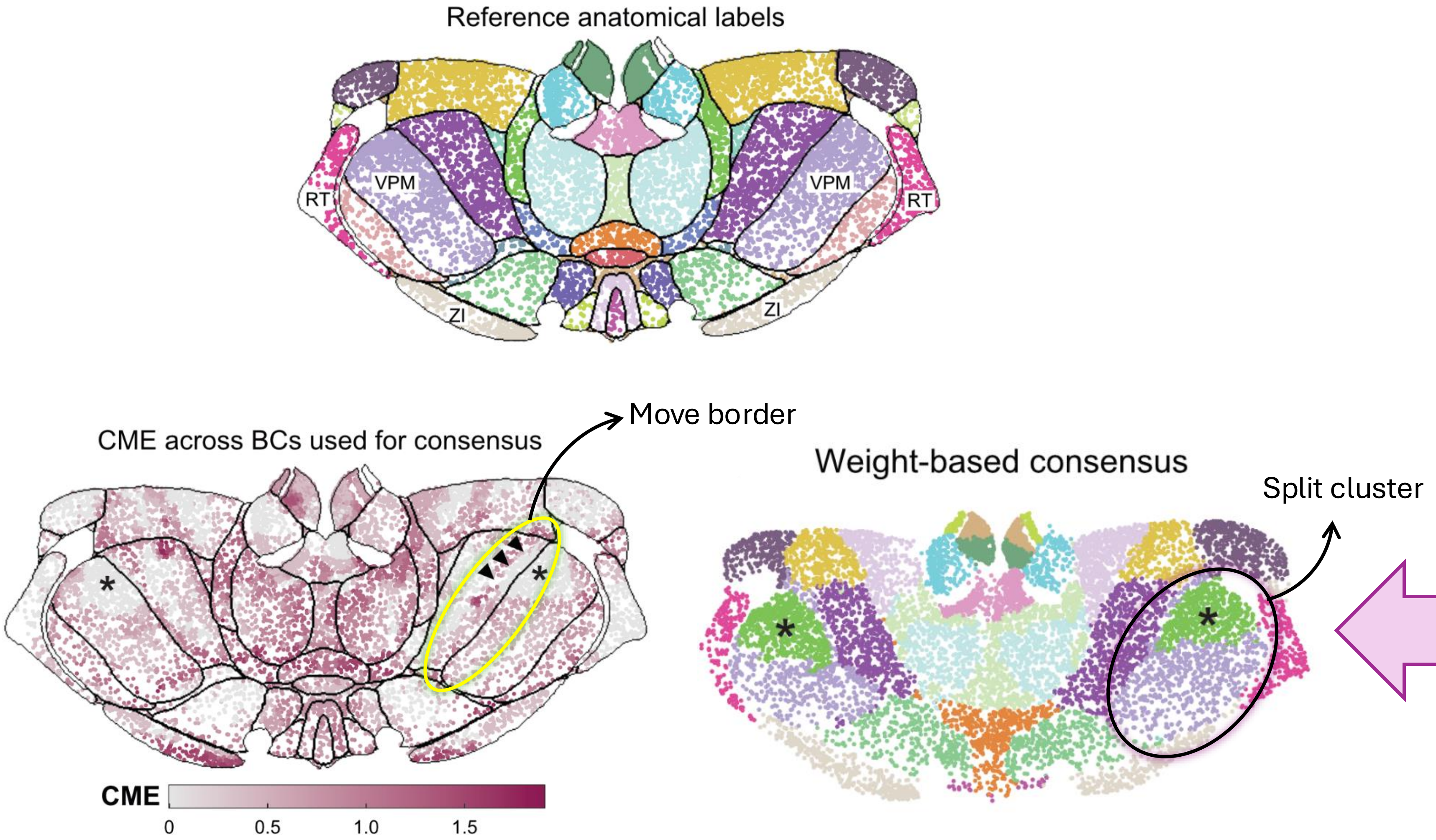
SCAN-IT
SOTIP
Stardust
...

Clustering as an **iterative** process between **methods** and **experts** with a **consensus**



- Maple
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- ...

bioRxiv

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Beyond benchmarking: an expert-guided consensus approach to spatially aware clustering

Jieran Sun, Kirti Biharie, Peiying Cai, Niklas Müller-Bötticher, Paul Kiessling, Meghan A. Turner, Søren H. Dam, Florian Heyl, Sarusan Kathirchelvan, Martin Emons, Samuel Gunz, Sven Twardziok, Amin El-Heliebi, Martin Zacharias, SpaceHack 2.0 participants, Roland Eils, Marcel Reinders, Raphael Gottardo, Christoph Kuppe, Brian Long, Ahmed Mahfouz, Mark D. Robinson, Naveed Ishaque

doi: https://doi.org/10.1101/2025.06.23.660861

bioRxiv

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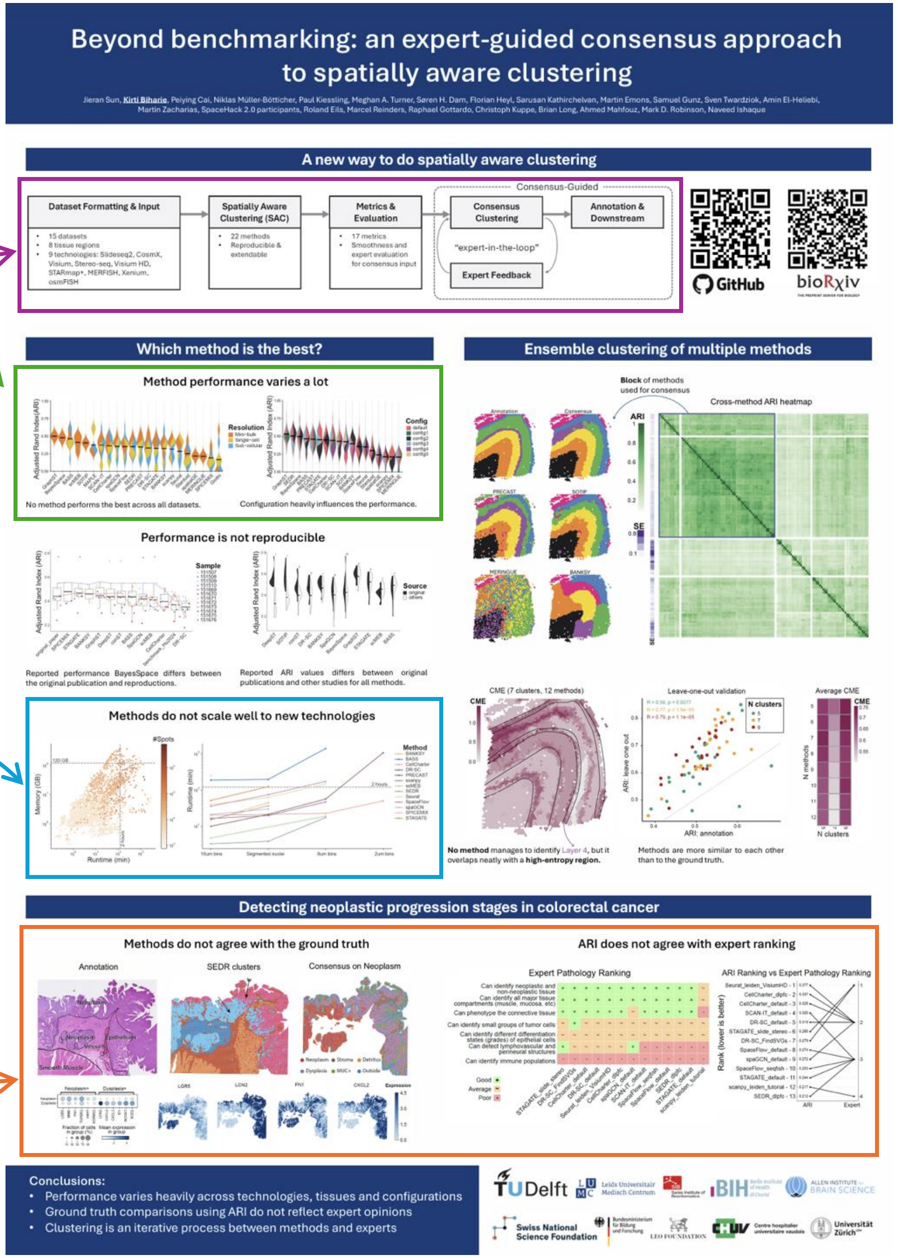
● ELIXIR-Germany 2nd BioHackathon ● 11-15 December 2023 ● Bielefeld, Germany ● Hybrid event ●

Extendable workflow

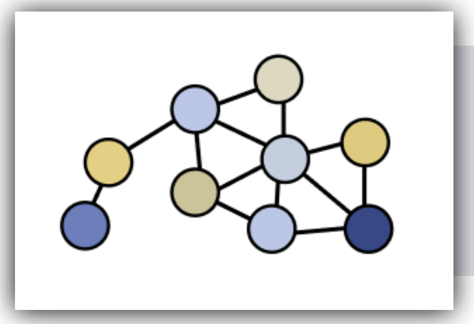
Configurations

Scalability

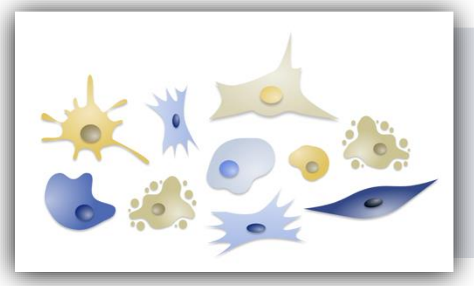
Application on Visium HD



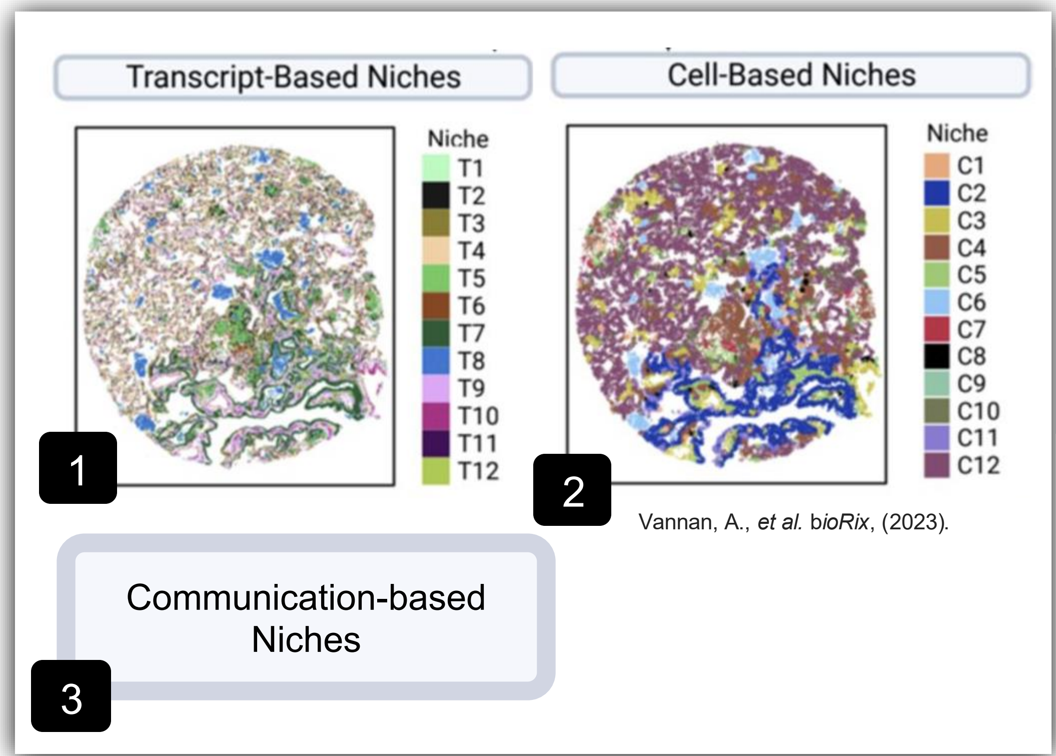
Summary: What are graphs and niches in spatial comics?



Graph: Often defined based on spatial proximity in µm or pixels



Niche: Cellular microenvironment often defined on a close to intermediate length scale



Vannan, A., *et al.* *bioRix*, (2023).

Niche definitions: Niches can be defined in various ways, currently there is no «one-fits-all» solution and optimal analysis steps depend on use-case/goal

Clinical relevance: Differences in spatial niches in disease cohorts can be liked to patient sub stratification and is useful for overall survival analysis.

Thank You!

a.mahfouz@lumc.nl
mahfouzlab.org